

PROVISIONAL PATENT APPLICATION

Rate-Control Devices and Systems Using a Resonantly Driven Ordered Medium: Qualified Rate-Control Module, Transportable Rate-Step Calibration Source, Physical-Clock Steering System, Payload-Region Rate-Control Enclosure, Process-Rate System, and Paired Differential-Rate System, and Methods of Operating, Qualifying, and Recalibrating Them

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This specification is complete and self-contained. Applicant's companion applications on support-force control (PROV-A), metrology instruments (PROV-C), and media and arrays (PROV-E) are nonessential background only; no material essential to written description, enablement, or the claims is incorporated from them. Every example under PROPHETIC EXAMPLES is prospective; no experiment, nonzero response, qualified unit, or controlled rate offset is represented as achieved.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is one of a coordinated series directed to applications of controlled angular-momentum coherence in condensed media. Companion drafts concern support-force control and calibrated support-force modules; transportation and lift systems; measurement instruments; energy storage and infrastructure; media and array architectures; information and winding-register devices; and completion-scheduled computation. This specification reproduces the product architecture, definitions, qualification apparatus, controls, and validation conditions essential to its own disclosure and does not rely on an unpublished companion for claim-essential matter.

FIELD

Rate-control devices and systems built around a resonantly driven ordered-medium module, and methods of operating them. More particularly: (G1) a **qualified rate-control module** whose controller commands, from a stored and independently validated empirical rate-response map, a setpoint fractional frequency-rate offset at a registered witness position and reports the achieved offset against a reference with uncertainty; (G2) a **transportable rate-step calibration source** that presents a commanded, map-derived fractional rate step to a receiving clock or comparator at a registered coupling position; (G3) a **physical-clock steering system** in which a clock's rate relative to a reference is steered by commanding the module rather than, or in addition to, electronic frequency correction; (G4) a **payload-region rate-control enclosure** in which plural modules and a spatial rate-response map establish a commanded

common fractional rate offset throughout a defined payload region; (G5) a **process-rate system** operable only after a non-clock rate process has been independently validated to follow the mapped response; and (G6) a **paired differential-rate system** operating two modules or regions at commanded differing setpoints from measured signs and maps. Each product is disclosed together with its packaging, map provenance, deployed operating method, factory qualification, runtime witnesses, service recalibration, safe-off behavior, and the boundary between the measured frequency-ratio response and any inferred physical interpretation. The multi-clock and force characterization apparatus is the **factory qualification stage** of these products, not the invention; it remains disclosed and claimed as a qualification method and apparatus. No nonzero proper-time effect, sign, magnitude, spatial law, or phase-to-rate transfer law is represented as experimentally established in this application.

BACKGROUND

1. Precision clocks detect relativistic rate differences and are also sensitive to ordinary environmental perturbations. GPS satellites carry a factory frequency offset of approximately -4.4647×10^{-10} so that the combined gravitational and motional rate difference in their specified orbit brings the transmitted signal into the system-rate convention. That practice is a calibration for a predicted orbital worldline; it does not locally actuate the proper time of a stationary clock package. Storage rings likewise establish time dilation for particles traveling on relativistic worldlines. These records provide calibration and comparison methods, not evidence that internally driven phase evolution in a stationary solid changes a body's proper-time rate.
2. A measured clock-frequency shift is not, by itself, a proper-time measurement. Static and alternating magnetic fields, electric fields, blackbody radiation, temperature, strain, vibration, acceleration, cavity pulling, line pulling, servo error, reference-distribution coupling, and other effects can shift an oscillator or atomic transition without changing the local proper-time rate. A directly driven material resonance is especially unsuitable as the sole time-rate witness because the drive intentionally changes that observable.
3. Extant clock steering is electronic: a local oscillator or atomic standard is corrected by a synthesizer offset, a servo bias, or a paper correction applied to its output, so that the indicated rate follows a reference. No extant product commands the rate of a clock, or of a co-located process, by commanding a physical driven medium adjacent to it under a stored, validated empirical response map. No extant transportable artifact presents a commanded fractional rate step of stated uncertainty to a receiving clock without an electronic frequency offset. Whether such products can be qualified depends on the measured response of a driven medium, which is presently unknown; the products disclosed here are therefore **conditional on qualification** and are constructed so that a null qualification result leaves a useful null-reference instrument rather than a false product.
4. Establishing the empirical response map requires a qualification apparatus combining: (i) a resonantly driven ordered medium; (ii) two physically dissimilar local clock technologies; (iii) a remote reference and continuous measurement of all three pairwise frequency ratios; (iv)

synchronized force, field, thermal, mechanical, drive-power, and reference-link measurements; (v) matched and swapped controls; and (vi) a preregistered classifier that accepts null and ordinary-artifact outcomes. The combination distinguishes a clock-specific perturbation from a common local rate response more effectively than a single driven clock or a force/clock correlation alone. That apparatus is disclosed here as the factory qualification stage.

5. The applicant's motivating hypothesis is that coherent phase advance of an aligned internal angular-momentum population may redistribute a body's rate budget and may therefore produce a common rate response in co-located clocks and processes. That hypothesis supplies a reason to build and qualify the products; it does not supply the presently unknown transfer relation, sign, magnitude, saturation law, spatial reach, or energy cost. Those are learned empirically at qualification and stored in the map. Unknown transfer mathematics is not a reason to narrow the product scope: the products are defined by their architecture, their stored validated map, and their operating method, not by a predicted law. A zero qualification result is a complete result.

Applicant's published exhibit corpus (nonessential background). Applicant maintains a public, commit-dated exhibit series presenting the underlying research program in didactic depth (the Normal Realism / POAMS exhibits, <https://cfmilam.github.io/nr-poams-exhibits/>). Those pages are applicant's own prior publications and are cited as nonessential background only. No material essential to enablement, written description, or the claims is incorporated from them.

SUMMARY

Six rate-control products and their operating methods are disclosed, all built on one **rate-control module (600)** (FIG. 1): an ordered medium; an alignment-field source; a coupling structure and locked drive chain applying an oscillating drive at the medium's located precession resonance; a registered witness position at which a witness clock, a customer clock, or a process is placed inside the module's mapped response region; a comparison port through which the witness is compared with a reference outside the mapped coupling domain; nuisance sensors; a controller; a calibration memory holding an **empirical rate-response map** with uncertainty, provenance, permissible interpolation region, runtime limits, and safe-off parameters; and a safe-off path. A unit or production class whose map contains at least one independently validated nonzero operating point is a **qualified rate-responsive module**; a unit whose map contains only null points is a **null reference**. Neither status presumes any response, sign, magnitude, additivity, or spatial law.

1. **G1 — Qualified rate-control module.** Accepts a fractional rate setpoint y_{set} for the witness position, confirms the setpoint lies within the permissible interpolation region, evaluates the map to obtain the drive state, applies the locked drive, regulates on the measured witness-to-reference ratio, monitors runtime witnesses, reports the achieved fractional offset with propagated uncertainty and validity state, and drives to safe-off on any validity failure.

2. **G2 — Transportable rate-step calibration source.** A module packaged with a receiving fixture at the registered coupling position, an internal witness clock, a comparison port, and a transport-sensitivity record; it presents a commanded map-derived fractional rate step to a receiving clock or comparator, with stated uncertainty, for checking comparator gain, step response, and link consistency without an electronic frequency offset.
3. **G3 — Physical-clock steering system.** A clock at the registered witness position, a reference input, a comparator, the module, and a steering controller that holds the clock's measured rate relative to the reference at a commanded value by commanding module setpoints within the map, with hand-off to conventional electronic steering, hold-over, and safe-off.
4. **G4 — Payload-region rate-control enclosure.** Plural modules arranged about a defined payload region, a **spatial rate-response map** established at qualification with witness clocks at registered interior, boundary, and exterior positions, an array allocation map, and a controller that commands a common fractional rate offset throughout the region within the map's spatial uncertainty. Additivity and reach are outputs of the map, not premises.
5. **G5 — Process-rate system.** Conditional on independent validation that a defined non-clock rate process at the witness position follows the mapped clock response within a stated tolerance; the system stores a process-rate map, commands a process-rate setpoint, and reports the achieved process-rate ratio with uncertainty. No process is represented as validated in this application.
6. **G6 — Paired differential-rate system.** Two modules or regions each with its own validated map, operated at commanded differing setpoints, including opposite signs only where each map contains validated operating points of that sign, with a differential witness comparison between the two positions and a common reference.

Layered scope (FIG. 2). Every output is disclosed at four layers: (L0) the raw ratio observable $y_{ij} = \Delta \ln(f_i/f_j)$ synchronized with the commanded drive state, always reportable and permitted to be zero; (L1) the validated rate-response map, entered only through the factory qualification gates G1–G10; (L2) controlled operation—setpoint control of the witness-to-reference ratio from the map by a qualified module or system in the *valid* state; and (L3) physical interpretation, in which a controlled common response is labeled a proper-time rate change only after independent non-clock or process-rate universality evidence and a supported physical account exist. No layer presumes the sign, magnitude, additivity, or spatial law of the response.

Three-tier architecture (FIG. 3). (i) The products and their deployed operating methods lead; (ii) factory qualification—the three-clock/force characterization apparatus, randomized control matrix, classifier, and gates G1–G10—establishes the empirical map and operating envelope once per unit or production class; (iii) validation and interpretation gates, runtime witnesses, service recalibration, and safe-off govern what a deployed unit may command and report. A purchaser operates a qualified module from its stored map without repeating the qualification laboratory; ordinary operators never re-run G1–G10.

BRIEF DESCRIPTION OF THE DRAWINGS

- FIG. 1 — Rate-control module (600): common product architecture with ordered medium, alignment-field source, coupling structure and locked drive chain, registered witness position, comparison port to a reference outside the mapped domain, nuisance sensors, controller, calibration memory, and safe-off path.
- FIG. 2 — Layered scope L0–L3: raw ratio observable, validated map behind gates G1–G10, controlled operation behind the runtime validity gate, and physical interpretation behind the non-clock universality gate.
- FIG. 3 — Qualification-to-deployment flow: factory qualification and map generation, classification as qualified rate-responsive module or null reference, deployed operation under runtime witnesses, service recalibration.
- FIG. 4 — Deployed operating method and validity-state machine: self-test, setpoint, region check, map evaluation, apply, regulate, monitor, report, zero checks, safe-off, recalibrate; states safe-off, self-test, valid, degraded, out-of-envelope, invalid.
- FIG. 5 — G2 transportable rate-step calibration source and G3 physical-clock steering system: product architectures with receiving fixture, internal witness, comparison port, transport record; steered clock, reference input, comparator, steering controller, and electronic-steering hand-off.
- FIG. 6 — G4 payload-region rate-control enclosure, G5 process-rate system, and G6 paired differential-rate system: plural modules about a payload region with spatial map and interior/boundary/exterior witnesses; process witness and process-rate map; two regions at differing setpoints with differential witness comparison.
- FIG. 7 — Factory qualification apparatus: driven medium, local clocks A and C, remote reference B (and second remote D), pairwise comparison, nuisance sensors, optional force accounts, and zero-valid classifier.
- FIG. 8 — Universality triangle: the required common local-to-remote response and unchanged local-to-local ratio, with clock-specific and remote/link-artifact alternatives.
- FIG. 9 — Randomized drive/control timing, absorbed-energy-matched states, blinded labels, discovery/confirmation separation, and synchronous extraction permitting a zero response.
- FIG. 10 — Joint clock, sample-force, source-reaction, field, thermal, mechanical, and drive channels; force correlation is corroborative rather than artifact-orthogonal.
- FIG. 11 — Classification ladder and validation gates G1–G10 from null or identified artifact through clock-specific residual to candidate common rate response.
- FIG. 12 — Empirical rate-response map admitting zero, the permissible interpolation region, and the conditional-control boundary.
- FIG. 13 — Spatial-response and array qualification mapping distance, shielding, orientation, inactive modules, and payload-region witness positions without presuming an enclosure effect (qualification of the G4 product).

- FIG. 14 — Redundant comparator network and independent-cycle residuals for detecting and, where observable, localizing clock or link faults.

DEFINITIONS (the applicant acts as his own lexicographer; these definitions govern the claims)

Portfolio operating terminology. "POAMS-native" and "direct-mode" identify the prophetic operating tier that applies the disclosed POAMS bookkeeping directly rather than requiring a preexisting nonzero empirical response map. "Commanded" and "selectable" identify controller- addressable input states and do not promise a nonzero output. "Independently validated" means reproduced under the disclosed qualification protocol against a separate reference; "empirically established" means recorded with uncertainty and provenance in a stored response map. Extant-framework labels introduced as such name measured phenomena without requiring the claims to adopt an associated entity ontology. Claim terms are operational and fixed by recited measurements.

1. **Ordered medium / internal angular-momentum population** — a material having an ensemble of constituent angular momenta measurable in aggregate, for example by magnetometry.
2. **Alignment field** — a measured static bias applied to establish or vary the population's alignment state and sense.
3. **Precession resonance** — an absorption resonance of the aligned population, located operationally by a measured extremum and linewidth (extant label: ferromagnetic resonance, FMR, for the reference YIG embodiment).
4. **Coherent phase advance** — drive-synchronous periodic advance of the measured collective precession phase during a continuous coherence interval. Cycle count alone is not presumed to be a retained state after drive termination.
5. **Local metrology frame** — a mechanically defined support or comparison region whose reference worldline, orientation, acceleration, temperature, and fields are measured. An extended body is not presumed to possess one unique proper time at all points.
6. **Clock** — an oscillator or transition-frequency standard with a documented sensitivity model and independently recorded output. The driven medium's precession output is a drive diagnostic and is not, by itself, a clock proving a common rate response.
7. **Physically dissimilar clocks** — two clocks based on different transitions or oscillator mechanisms and having materially different magnetic, electric, thermal, strain, optical, and servo sensitivity coefficients. A hyperfine standard and an optical atomic or ion standard are preferred for the strongest universality test; atomic, crystal, and optical-resonator combinations may serve earlier artifact-screening embodiments.
8. **Remote reference** — a clock outside the mapped drive, field, thermal, mechanical, and electrical-coupling domain, connected by a characterized one-way or preferably reciprocal two-way comparison link. For the strongest common-response test, a second independently operated remote

reference D, or an equivalent registered remote-clock and reference-link swap, distinguishes a change confined to remote clock B from a change common to local clocks A and C.

9. **Fractional response** — for clocks i and j having measured frequencies f_i and f_j , the baseline-referenced log-ratio change $y_{ij} = \Delta \ln(f_i/f_j)$. For local clocks A and C and remote reference B, the controller records y_{AB} , y_{CB} , and y_{AC} .
10. **Universality triangle** — the simultaneous condition that y_{AB} and y_{CB} have the same drive-correlated value within combined uncertainty while y_{AC} remains zero within its uncertainty. It is a necessary discriminator for a common local rate response, not sufficient proof of proper time.
11. **Clock-specific response** — a change observed in one clock technology or proportional to a known clock sensitivity coefficient and not satisfying the universality triangle.
12. **Candidate common rate response** — a nonzero universality-triangle response that also survives the validation gates below and independent replication. The term is deliberately interpretation-neutral.
13. **Drive-state vector ($\mathbf{x}$)** — independently measured variables including alignment-field magnitude and sense, drive frequency and phase, incident/reflected/transmitted and absorbed power, pulse timing and duty, loaded quality factor, resonance mode and linewidth, aligned population or magnetization, material volume, witness/clock positions, temperature, pressure, orientation, shielding state, and time.
14. **Nuisance-state vector ($\mathbf{n}$)** — measured clock sensitivities, local and remote fields, field gradients, drive leakage, blackbody environment, temperature, strain, vibration, acceleration, pressure, cavity pulling, servo state, reference distribution, ground coupling, and link delay/nonreciprocity.
15. **Absorbed-energy-matched control** — a detuned, phase-scrambled, sham, or reference-body state adjusted to match absorbed-energy and temperature history, using incident, reflected, transmitted, and preferably calorimetric measurements. Equal source power alone is insufficient.
16. **Empirical rate-response characteristic $\mathbf{R}$; empirical rate-response map** — a preregistered estimate of the measured frequency-ratio and optional force responses as a function of $\mathbf{x}$, $\mathbf{n}$, and control state, together with uncertainty, validity range, hysteresis, bandwidth, saturation, spatial dependence, and null residuals. $\mathbf{R}$ may be identically zero. The *map* is the stored form of $\mathbf{R}$ in a calibration memory: a set of validated operating points (drive state $\rightarrow$ fractional response at a registered witness position, with covariance), the permissible interpolation region among them, provenance, runtime limits, and safe-off parameters.
17. **Validated drive-correlated frequency response** — a nonzero response that survives applicable artifact gates, multiplicity correction, swaps, a frozen blinded confirmation, and independent replication. It may remain clock-specific.
18. **Validated candidate common rate response** — a validated response that additionally satisfies the universality triangle and the clock/link systematic budget. It is not represented as an achieved proper-time response.

19. **Spatial rate-response map** — a measured response as a function of witness position, distance, orientation, shielding, module count, module state, and drive condition, including inactive and sham-module controls, with uncertainty at each registered position. Additivity across modules and reach beyond the measured positions are not presumed.
20. **Conditional empirical-control mode; setpoint mode** — operation enabled only after a validated response has been mapped, and only inside its permissible interpolation region with continuing validity monitoring and automatic return to safe-off or a measurement state.
21. **Independent replication** — repetition on a separately assembled apparatus and preferably by an independent laboratory under a frozen protocol. A held-out repeat on the same apparatus is confirmatory evidence but is not, by itself, independent replication.
22. **Registered witness position** — a mechanically defined location and orientation within the module's mapped response region, fixed by a fixture or interface, at which the qualification witness clocks were compared and at which a deployed witness clock, customer clock, receiving clock, or process is placed. The map is valid only at registered positions and within the stated positional tolerance.
23. **Witness clock** — a clock, of a technology whose sensitivity coefficients are recorded in the map, placed at a registered witness position in a deployed unit and compared with a reference through the comparison port to report the achieved fractional offset. A witness clock is a runtime witness of the L2 quantity; it is not, by itself, evidence of L3.
24. **Reference** — for a deployed unit, a clock or timescale outside the module's mapped coupling domain, connected through the comparison port by a characterized link; for qualification, remote references B and D of Definition 8.
25. **Comparison port** — the interface (optical or electrical frequency-comparison link, two-way time-transfer link, or common-view link) through which the witness or steered clock is compared with the reference, with recorded link delay, nonreciprocity, and status.
26. **Fractional rate setpoint (y_{set})** — a commanded change in the witness-to-reference log-ratio relative to the drive-off baseline at a registered witness position, with sign, or a time-varying profile of such values within stored bandwidth and slew limits; for G4, a common value commanded throughout a defined payload region; for G5, a process-rate ratio; for G6, a pair of values at two positions.
27. **Permissible interpolation region** — the region of drive and environmental state, and of setpoint value, within which the map has validated operating points or held-out points confirming the interpolation model within stored tolerance. Extrapolation beyond the region is not permitted.
28. **Runtime witnesses** — the reduced set of sensor readings monitored during deployed operation and compared with the runtime limits: at least the witness-to-reference ratio and its uncertainty, comparison-link status, alignment-field magnitude and sense at the medium and at the witness position, drive lock state, incident/reflected/absorbed power and duty, medium and witness temperature, three-axis field and gradient at the witness position, drive leakage at the witness position, vibration/acceleration, pressure, and witness-clock servo state.

29. **Safe-off** — the stored state to which the controller drives the unit on a validity failure, fault, or command: oscillating drive off; alignment field to a stored safe magnitude and sense (or unchanged in a fixed-bias unit); witness comparison continued and logged; last valid record preserved; recalibration flag set; for G3, hand-off of clock steering to the conventional electronic steering path or hold-over.
30. **Validity states** — the controller-reported states *self-test*, *valid* (setpoint mode permitted), *degraded* (L0 and L1 reported; setpoint held or reduced; update required), *out-of-envelope* (a runtime witness lies beyond a runtime limit or the setpoint lies outside the permissible region; setpoint inhibited), *invalid* (map check or gate failure; recalibration flagged), and *safe-off* (Definition 29), as shown in FIG. 4.
31. **Qualified rate-responsive module; qualified product class; null reference** — a unit, or a production class represented by qualified units, whose calibration memory holds an empirical rate-response map containing at least one independently validated nonzero operating point under Definitions 17–18 and gates G1–G9 (and G10 for any spatial claim), its uncertainty, provenance, permissible interpolation region, runtime limits, and safe-off parameters, is a *qualified rate-responsive module* (or *class*). A unit whose map contains only null operating points is a *null reference*: it accepts test-state commands and reports L0 quantities but has no enabled setpoint mode. Neither status presumes any response, sign, magnitude, additivity, or spatial law beyond the validated points.
32. **Provenance record** — the qualification identifier, apparatus and laboratory identifiers, dates, frozen protocol version, clock technologies and sensitivity coefficients used, gate results G1–G10 with uncertainties, replication record, held-out residuals, and the unit or class identifier to which the map applies.
33. **Service recalibration** — a reduced re-qualification performed at the service interface: drive-off and detuned zero checks, a witness-clock ratio check against the reference at one or more validated operating points, comparison-link reciprocity check, and, where the check exceeds the stored tolerance, return of the unit to full qualification.
34. **Process-rate map** — for G5, the measured relation, established under the qualification control matrix with the process at the registered witness position, between the commanded drive state, the co-located witness-clock response, and an independently measured rate index of a defined non-clock process (for example a reaction-rate, decay-count, oscillator-independent physical-clock, or biological-marker index), with uncertainty. The process-rate map exists only if the process response was validated independently of any clock response; it is not inferred from the clock map.
35. **Array allocation map** — for G4 or G6, the measured relation between a set of per-module drive states and the fractional response at each registered witness position of the assembled plurality, established under sequential, in-phase, phase-scrambled, and inactive-module qualification states, with uncertainty and cross-talk terms. The sum of individual module responses is not presumed.
36. **Interpretation boundary** — the stated separation, carried in every report, between the L2 quantity (achieved witness-to-reference fractional offset at a registered position, with uncertainty) and any

L3 statement (proper-time rate change, process aging change). L3 is reported only where the provenance record contains the independent non-clock or process-rate universality evidence of Definition 34 and a supported interpretation.

SCIENTIFIC FOUNDATION AND OPEN TRANSFER QUESTION

T1 — Established comparison physics. Proper time is associated with a worldline. Motion and gravitational potential produce well-tested differences among clock readings when their worldlines and transfer conventions are defined. GPS factory frequency offsets and storage-ring lifetime dilation are accepted calibration records for predicted worldlines. They do not establish the proposed stationary-medium effect and are not relied upon as its enabling mechanism.

T2 — Clock indication versus common rate. An unperturbed transition frequency can serve as a clock only after known shifts are evaluated. The SI second is defined through the unperturbed cesium-133 hyperfine transition; that definition is a measurement convention and establishes neither the applicant's rate-inventory hypothesis nor any conversion between an internal phase count and proper time—any such conversion is a hypothesis to be tested, not metrological precedent. The qualification local-clock pair is therefore chosen to have dissimilar sensitivity coefficients. With $y_{ij} = \Delta \ln(f_i/f_j)$, a candidate common response requires approximately

$$y_{AB} = y_{CB} = \delta \text{ and } y_{AC} = 0$$

within the combined uncertainty budget. A common reference-link error or a change confined to remote clock B can mimic the first two equalities. Reciprocal links, link monitors, position and remote-reference swaps, and preferably a second independently operated remote reference test those alternatives. A common magnetic, thermal, or mechanical environment can also perturb both local clocks; nuisance measurements and deliberately injected positive controls test those alternatives.

T3 — Motivating rate-budget hypothesis. Energy and frequency obey $E = h\nu$. The applicant's POAMS interpretation treats organized internal turning as a rate ledger and hypothesizes that coherent phase advance may redistribute that ledger. A convenient geometric mnemonic is $t^2 = \tau^2 + (s/c)^2$. For translational motion, s has an ordinary worldline meaning. This application does not assume that an internal phase angle supplies the same s , does not assign an unstated mechanical radius, and does not claim that the mnemonic derives the proposed stationary-medium response.

T4 — Open transfer relation, and why it does not narrow the products. Before experiment, no dimensionally complete law is known for

$R = R(B_0, B_1, \omega, \text{phase}, Q_L, P_{\text{abs}}, \text{duty}, N_{\text{aligned}}, V, T, \Gamma, \text{geometry}, \text{shielding}, \text{position}, \text{time}, n).$

The sign, magnitude, saturation, bandwidth, persistence, spatial reach, and energy cost are measured outputs; $R = 0$ is allowed. Drive frequency or cycle count alone does not predict a fractional clock response. The products of this application do not depend on a closed-form R : each is operated from the stored map of validated operating points and their permissible interpolation region, and each is qualified

by measurement. A later nonzero result may support a fitted constitutive relation, but the search data and held-out confirmation data are kept separate, and no fitted relation extends the permissible region beyond validated or held-out-confirmed points.

T5 — Force channel. A simultaneous support-force response can strengthen causal diagnosis, but clock and force artifacts are not disjoint. Magnetic gradients, heating, strain, vibration, reference motion, and electrical coupling can affect both. The force channel is corroborative and is interpreted only with sample-force, source-reaction, field-gradient, acceleration, and thermal measurements.

T6 — Scaling, energy, and the interpretation boundary. No metrology-to-enclosure scale law is assumed. A controller interpolates only within a validated measured map. Extrapolation in magnitude, module count, distance, or payload volume requires new registered qualification tests. Any claimed universal-rate or spatial effect must carry an energy and momentum account. A proper-time interpretation requires evidence of universality beyond one clock transition and beyond clocks—an independently measured non-clock rate process. This disclosure supplies the products, their operating methods, the qualification apparatus, and the decision gates; it does not supply a working proper-time result, and every product report carries the interpretation boundary of Definition 36.

T7 — Comparator-network identity. Pairwise frequency ratios multiply algebraically around a closed simultaneous comparison loop. Loop closure tests comparator and link consistency; it does not prove that physical actuator stages multiply, and no product in this application stages modules multiplicatively. Fault localization requires redundant independent cycles and an observable fault model, not one loop alone.

THE RATE-CONTROL MODULE (600) — COMMON PRODUCT ARCHITECTURE (FIG. 1)

Ordered medium (602). The reference body is a room-temperature polished single-crystal YIG sphere, nominally 10 mm in diameter, whose actual mass, dimensions, magnetization curve, saturation state, resonance spectrum, linewidth, and phase response are measured and recorded in the provenance record. Alternatives are other garnets, ferrites, and engineered or composite media admitted under E10 after their own qualification. The medium is carried in a **specimen holder (603)** of low-loss nonmagnetic material (quartz, sapphire, PTFE, alumina).

Alignment-field source (604). A separately supported electromagnet, permanent-magnet yoke, or coil-and-yoke assembly producing a static field of commanded magnitude and sense at the medium (reference: a C-frame electromagnet of the GMW-5403 class supplying approximately 0.196 T at approximately 20 A and an 80-mm gap for qualification; a permanent-magnet yoke with a trim winding for deployed units, as the unit's map dictates). The field and its gradient at the medium and at the registered witness position are mapped and stored; the witness-position field is a runtime witness because a witness clock's own field sensitivity is a recorded nuisance coefficient.

Coupling structure and locked drive chain (606, 608). A coupling structure (mapped rectangular TE-mode cavity, loop-gap resonator, re-entrant resonator, waveguide, or coil) places the medium at a mapped antinode of the oscillating drive's magnetic component. The reference structure is a rectangular TE₁₀₂

cavity with internal dimensions approximately $47.55 \text{ mm} \times 22.15 \text{ mm} \times 64.77 \text{ mm}$ for a 5.600-GHz reference mode. The drive chain comprises a synthesizer, modulator, power amplifier, directional coupler, incident/reflected and, for a two-port structure, transmitted-power detection, and the resonance-lock servo. Loaded Q, ring-up, ring-down, absorbed power, and medium temperature are measured rather than inferred from source power. Drive leakage at the witness position is measured and is a runtime witness.

Registered witness position and fixture (610). A mechanically defined location and orientation, within the module's mapped response region and outside the coupling structure, fixed by a fixture, rail, or standard interface so that a witness clock, customer clock, receiving clock, or process occupies the same position and orientation as the qualification witness clocks within the stored positional tolerance. Shielding between the medium/coupling structure and the witness position (magnetic, thermal, vibration, electrical) is part of the qualified configuration and is recorded; changing it invalidates the map.

Witness clock (612) and comparison port (614). A witness clock of a technology whose sensitivity coefficients are stored in the map is placed at the witness position (or the customer's clock serves as witness in G3). The comparison port carries the witness output to a reference (616) outside the mapped coupling domain through a characterized link with recorded delay, nonreciprocity, and status; a reciprocal or redundant link is preferred. The module records $y_W = \Delta \ln(f_{\text{witness}}/f_{\text{reference}})$ continuously.

Nuisance sensors (618). Three-axis field and gradient at the medium and at the witness position; drive leakage at the witness position; medium, holder, coupling-structure, witness, and ambient temperature; blackbody environment at the witness; strain at supports; vibration and three-axis acceleration; pressure; witness-clock servo state; comparison-link monitors; incident/reflected/transmitted/absorbed power and duty.

Controller (620) and calibration memory (622). The controller executes the deployed operating method of FIG. 4: it loads and authenticates the map, accepts a fractional rate setpoint, confirms the permissible region, evaluates the map, applies and locks the drive, regulates on the measured witness ratio, monitors the runtime witnesses against the runtime limits, reports the achieved offset with propagated uncertainty and the validity state, and drives to safe-off on any failure. The calibration memory holds the map (Definition 16), provenance record (Definition 32), permissible interpolation region, runtime limits, hysteresis and thermal coefficients, zero-check history, recalibration schedule, and safe-off parameters. Every report carries the interpretation boundary (Definition 36).

Safe-off path (624). A hardware-enforced drive interrupt and a stored alignment-field safe state, reachable from every controller state and on controller fault, with the witness comparison continued and logged through the transition.

Energy and momentum account. The module records incident, reflected, transmitted, dissipated, stored, mechanical, and thermal energy at each operating point; a qualified unit's map carries this account, and no operating point is stored without it.

DEPLOYED OPERATING METHOD (FIG. 4)

The deployed operating method is performed by a qualified rate-responsive module or by a system of G2–G6 whose calibration memory already holds a validated map. Its steps are: (a) *Power-on and self-test*. Load the map; verify that the provenance record matches the unit or class identifier and that the witness clock technology and position match the map; verify comparison-link status and reciprocity; execute a drive-off zero check and a detuned absorbed-energy-matched check of y_W against the stored baseline; enter *valid* only if all pass, otherwise *invalid* or *safe-off*. (b) *Receive setpoint*. Accept a fractional rate setpoint y_{set} (Definition 26) from an operator, a supervisory controller, or a stored profile. (c) *Check region*. Confirm that y_{set} and the current environmental state lie within the permissible interpolation region and runtime limits; otherwise inhibit and report *out-of-envelope*. (d) *Determine mapped state*. Invert or interpolate the map to obtain the drive state (alignment-field magnitude and sense, frequency, phase, power, pulse width, duty, burst pattern) and its predicted y_W with uncertainty. (e) *Apply*. Ramp the alignment field (if commanded) in a settled block and apply the locked drive within the stored slew and bandwidth limits. (f) *Regulate*. Feedforward from the map; close the loop on the measured y_W with the stored gain bounds, subject to the witness clock's and link's measured noise so that the regulation bandwidth never exceeds the bandwidth at which y_W is resolved to the stored tolerance. (g) *Monitor*. On each control cycle compare the runtime witnesses with the runtime limits and test that the measured y_W change is not accounted for by the recorded witness-clock sensitivity coefficients acting on the measured field, gradient, leakage, temperature, or servo change at the witness position (the runtime artifact test). (h) *Report*. Output the achieved fractional offset with propagated uncertainty, the validity state, and the interpretation boundary statement. (i) *Zero checks*. At the stored interval interleave drive-off and detuned zero checks; drift beyond tolerance transitions to *degraded*, shrinks the permissible region, or flags recalibration. (j) *Safe-off*. On a runtime-limit exceedance, a runtime artifact-test failure, loss of drive lock, comparison-link failure, or fault, transition to *safe-off* and set the recalibration flag. (k) *Recalibrate*. Perform service recalibration (Definition 33) or return the unit to full qualification. The operator of a deployed unit does not perform the qualification protocol of E1–E6; gates G1–G10 are executed once at qualification and again only on service recalibration or return to the factory.

LAYERED SCOPE (FIG. 2)

- **L0 — raw ratio observable.** y_{ij} synchronized with the commanded drive state, with statistical uncertainty from the recorded intervals and drift model; at qualification y_{AB} , y_{CB} , y_{AC} (and y_{AD} , y_{CD}); in deployment y_W . L0 is always reportable; it may be zero.
- **L1 — validated rate-response map.** The map-corrected fractional response with uncertainty at a recorded drive and environmental state and registered witness position, reportable when that state lies inside the permissible region. Gates G1–G9 (and G10 for any spatial point) stand between L0 and L1: no operating point is a validated point of the map until it has passed them.

- **L2 — controlled operation.** Setpoint control of y_W from the map by the module of FIG. 1 or a system of G2–G6. The runtime validity gate (Definition 30) stands between L1 and L2: a deployed unit commands only validated or permissibly interpolated points and only in the *valid* state. An L2 report states an achieved witness-to-reference fractional offset; it does not state a proper-time change.
- **L3 — physical interpretation.** A controlled common response is labeled a proper-time rate change, or a process-rate (aging) change, only where the provenance record contains independent non-clock or process-rate universality evidence (Definition 34), the G10 spatial map where the claim is spatial, and a supported physical account. The non-clock universality gate stands between L2 and L3. No L3 statement is made in this application.

QUALIFICATION, CALIBRATION, AND DEPLOYMENT (FIG. 3)

Factory qualification versus deployed operation. Qualification is performed once per unit or production class with the full three-clock/force apparatus and control matrix of E1–E6; deployed operation uses the stored map, the runtime witnesses, and the reduced self-tests of the deployed operating method. A deployed unit never commands an operating point its map does not support.

Empirical-map procedure. (a) *Acquire*: with the module's medium, coupling structure, alignment source, shielding, and witness fixture in their production configuration, execute the randomized sequence of drive and control states of E3 across the intended envelope, with local clocks A and C at the registered witness position(s) and remote references B and D, recording all synchronized channels. (b) *Fit*: estimate $R(T4)$ as a function of drive state with a preregistered model class (lookup with interpolation, polynomial, or physics-agnostic regression), with covariance, hysteresis, and time constants, alongside the nuisance model. (c) *Validate*: classify each candidate operating point under E4 as null, identified artifact, clock-specific residual, or common residual; apply gates G1–G8 to any nonzero common residual; obtain independent replication (G9); apply G10 to any point claimed at a position other than the reference witness position; test the fitted map on held-out operating points. (d) *Store*: write the map, covariance, provenance, permissible interpolation region, runtime limits, energy account, and safe-off parameters to the calibration memory, and record the qualification status—qualified rate-responsive module if at least one nonzero operating point passed G1–G9, null reference otherwise. (e) *Ship and operate*: the deployed operating method of FIG. 4 with a witness clock of a technology recorded in the map. (f) *Update*: runtime zero and reference checks are logged; drift beyond tolerance shrinks the region or triggers recalibration.

Production-class qualification. A class map is established from a plurality of qualified units of common design. An additional unit is accepted under the class map only after an acceptance test at one or more validated operating points with a witness clock and reference agrees with the class map within a stored tolerance; a unit that does not agree is qualified individually or as a null reference.

Array and spatial qualification (G4, G6). For a plural-module system the assembled plurality is qualified under E8 with witness clock pairs at registered interior, boundary, and exterior positions and sequential, in-phase, phase-scrambled, inactive, sham, distance, and shielding states, producing the spatial rate-response map (Definition 19) and array allocation map (Definition 35). A payload-region setpoint mode is enabled only for a region throughout which G10 passed.

Process qualification (G5). The defined process is placed at the registered witness position beside witness clock A and C and its rate index is measured under the same randomized control matrix; the process-rate map (Definition 34) is stored only if the process index follows the commanded state under the same gates, independently of the clock map, with replication (G9).

What qualification does not presume. Sign, magnitude, linear scaling with population or power, additivity across modules, spatial reach, persistence after drive termination, any process-rate or aging effect, and any proper-time interpretation are outputs of measurement; each remains absent from the map until measured and validated. A null qualification result produces a useful null reference and an upper bound.

DETAILED DESCRIPTION — THE PRODUCT BRANCHES (FIGS. 1, 4, 5, 6)

Each branch below is self-contained and conditional: it is operable as described only for a qualified rate-responsive module or class whose stored map supports the recited setpoints. A unit whose qualification returned a null is shipped, if at all, as a null reference with the setpoint mode disabled. Nothing in any branch presumes a response, sign, magnitude, additivity, spatial law, or physical interpretation; each of those is read from the unit's own map.

G1 — Qualified rate-control module (650; FIGS. 1 and 4). *Product.* The module of FIG. 1 packaged as a bench or rack unit: an evacuable or temperature-controlled housing carrying the medium, holder, alignment source, coupling structure, and drive chain on a reaction frame; a shielded witness bay presenting the registered witness position through a standard clock-mounting interface; comparison-port connectors (optical fiber and coaxial) with link monitors; a nuisance-sensor harness; controller and calibration memory; a setpoint interface (front panel, remote command, stored profile); a hardware safe-off interrupt; and a report output. *Map provenance.* The map is written only by the factory qualification of E1–E6 with local clocks A and C at the witness bay and remote references B and D; the provenance record travels with the unit and is authenticated at power-on. *Setpoint control.* The controller performs steps (a)–(k) of the deployed operating method. The setpoint is a fractional offset y_{set} of the witness clock relative to the reference; the report is the achieved y_{W} with uncertainty. *Runtime witnesses and artifact test.* Definition 28; the runtime artifact test of step (g) uses the witness clock's stored sensitivity coefficients and the measured field, gradient, leakage, temperature, and servo state at the witness bay, so that an offset explained by an ordinary witness-clock perturbation is reported as an artifact and the unit transitions to *invalid*. *Service recalibration and safe-off.* Definitions 33 and 29. *Interpretation boundary.* The unit reports "achieved witness-to-reference fractional offset at registered position"; it does not report

a proper-time change. A second witness clock of a dissimilar technology may occupy a second registered position to report a runtime universality check ($y_{W1} = y_{W2}$ within uncertainty), which strengthens the L2 report and is a necessary input to any later L3 statement.

G2 — Transportable rate-step calibration source (660; FIG. 5, left). *Product.* A ruggedized transport case containing the module (600) with a fixed-bias permanent-magnet alignment source and trim winding, an internal witness clock (612) at a first registered position, a **receiving fixture (662)** presenting a second registered position to a receiving clock, or a comparison-port adapter presenting the receiving clock's comparator input directly, transport sensors (shock, tilt, temperature, field exposure) with a **transport record (664)**, comparison-port connectors, controller, calibration memory, and report output. *Function.* On command, the source steps between the drive-off baseline and a validated operating point, presenting to the receiving clock or comparator a fractional rate step Δy of stated value and uncertainty derived from the map at the receiving position, without any electronic frequency offset in the receiving clock's path. The receiving system's indicated step is compared with the commanded step to check comparator gain, step response, settling time, and link consistency. *Map provenance and transport.* The map is established at qualification with witness clocks at both registered positions; transport sensitivity is qualified by pre- and post-transport ratio checks at validated operating points, and a stored transport limit (shock, field exposure, temperature excursion) beyond which the unit reports *invalid* until re-checked. *Runtime witnesses.* The internal witness clock's y_W is compared in real time with the commanded step; a deviation beyond tolerance invalidates the presented step. *Traceability.* The step is traceable to the qualification remote references through the provenance record; it is not traceable to the SI second by any presumed phase-to-rate law. *Safe-off, recalibration, boundary.* As for G1. The source certifies "commanded fractional step at registered position"; it does not certify a proper-time step.

G3 — Physical-clock steering system (670; FIG. 5, right). *Product.* A steered clock (672)—the customer's atomic standard, ion or optical clock, or oscillator, of a technology whose sensitivity coefficients are recorded in the map—mounted at the registered witness position; a reference input (674) from an external timescale or remote clock through the comparison port; a comparator (676); the module (600); a **steering controller (678)** that commands module setpoints; a conventional **electronic steering path (679)** (synthesizer offset or servo bias) in parallel; hold-over logic; and a report output. *Operating method.* The steering controller measures $y_{\text{steered}}/\text{reference}$, computes the steering correction required to hold the steered clock's rate at a commanded value relative to the reference (including zero), partitions the correction between the module setpoint (within the permissible region and the map's bandwidth) and the electronic path (for any residual and for all correction outside the region), and reports which portion of the correction was physical and which electronic, each with uncertainty. *Hand-off and hold-over.* On any validity failure the module drives to safe-off and the full correction transfers to the electronic path or to hold-over without interruption of the clock output; the transfer is logged. *Runtime witnesses.* Definition 28, plus the steered clock's own systematic monitors (field, temperature, blackbody, servo) so that the artifact test of step (g) can distinguish a module-commanded rate change from an ordinary perturbation of

the steered clock. *Interpretation boundary.* The system reports "physically steered fractional rate relative to reference"; whether the steered clock's proper time was changed is an L3 statement not made by the product. A second co-located dissimilar witness clock may report the runtime universality check.

G4 — Payload-region rate-control enclosure (680; FIG. 6, top). *Product.* A plurality of modules (600) arranged about a defined **payload region (682)**—a shelf, chamber, container, or vehicle bay—on a common frame with mapped shielding; witness clocks at registered interior, boundary, and exterior positions (684); a common reference through the comparison port; an **array controller (686)** holding the spatial rate-response map (Definition 19) and array allocation map (Definition 35); and a report output. *Setpoint control.* The array controller accepts a common fractional rate setpoint for the payload region, confirms that the region and setpoint lie within the spatially validated permissible region, evaluates the array allocation map to obtain per-module drive states, applies them, regulates on the interior witnesses, and reports the achieved offset at each registered position with uncertainty and the spatial uniformity residual. *Qualification.* The spatial map is established by E8 and gate G10 with witness clock pairs at every registered position and inactive, sham, sequential, in-phase, phase-scrambled, distance, orientation, and shielding states; the region is defined as the set of positions at which G10 passed. Additivity across modules and reach beyond the registered positions are read from the map, never presumed. *Runtime witnesses and safe-off.* Interior and exterior witnesses; any interior witness disagreement beyond tolerance, or any exterior witness change beyond the stored exterior limit, drives the array to safe-off. *Interpretation boundary.* The enclosure reports "achieved common fractional offset of the interior witnesses at registered positions"; it does not report that the payload's proper time or aging has changed. A payload process effect is a G5 matter.

G5 — Process-rate system (690; FIG. 6, middle). *Product.* A G1 module or G4 enclosure whose witness position(s) additionally carry a defined non-clock **process (692)** and an independent **process-rate sensor (694)** reading its rate index; a calibration memory holding a process-rate map (Definition 34) in addition to the clock map; and a controller that accepts a process-rate setpoint, evaluates the process-rate map, applies the module, regulates on the co-located witness clock, and reports the achieved process-rate ratio with uncertainty from the process-rate sensor. *Condition.* This branch is operable only if factory qualification has independently validated, under the full control matrix and gate G9, that the defined process's rate index follows the commanded drive state, and the process-rate map records that validation. No process is represented as validated here; perishable, chemical, biological, and other process-rate applications are prophetic until such a map exists for the specific process. *Runtime witnesses.* Definition 28, plus the process-rate sensor and the process's own environmental monitors (temperature, humidity, illumination, atmosphere), so that an ordinary environmental change in process rate is reported as an artifact. *Interpretation boundary.* The system reports "achieved process-rate ratio for the defined process at registered position"; it does not generalize to other processes or to proper time. A validated process-rate map is, however, the non-clock universality evidence that Definition 36 requires before any L3 statement.

G6 — Paired differential-rate system (695; FIG. 6, bottom). *Product.* Two modules (600a, 600b) or two G4 regions, each with its own validated map, mounted on a common frame with a mapped separation and shielding; witness clocks at each registered position; a common reference; a **differential comparator**

(697) measuring $y_a - y_b$ directly between the two witnesses and each against the reference; and a **differential controller (698)** commanding $y_{set,a}$ and $y_{set,b}$. *Condition.* Setpoints of differing value are commanded only where each map contains validated operating points at the commanded values; setpoints of opposite sign are commanded only where each map contains validated operating points of the commanded sign. The availability of either sign is a qualification result, not a design premise. The cross-coupling of module a on witness b (and vice versa) is measured at qualification and stored in the array allocation map. *Uses (prophetic).* Differential-rate calibration of comparators and links (a commanded differential step without an electronic offset), differential aging or process tests under G5 maps, and common-mode rejection of reference drift in the reported differential. *Runtime witnesses, safe-off, boundary.* As for G1 applied to each module, plus the differential comparator; safe-off of either module safe-offs both. The report is the achieved differential fractional offset between registered positions.

DETAILED DESCRIPTION — FACTORY QUALIFICATION APPARATUS AND PROTOCOL (FIGS. 7–14)

The following embodiments constitute the factory qualification stage. They are performed once per unit or production class and at service recalibration. They are the enabling path from an unknown response to a stored validated map; they are not the product.

E1 — Self-contained driven-medium reference assembly (FIG. 7). The enabled reference specimen is the medium (602) of the module under qualification: a room-temperature single-crystal YIG sphere, nominally 10 mm in diameter, whose actual mass, dimensions, magnetization curve, saturation state, resonance spectrum, linewidth, and phase response are measured. The mapped rectangular TE102 cavity seed has internal dimensions approximately 47.55 mm × 22.15 mm × 64.77 mm for a 5.600-GHz reference mode and places the specimen at a transverse oscillating magnetic-field antinode. Loop-gap, re-entrant, waveguide, coil, and mapped off-axis structures are alternatives after their loaded field distributions are measured. A separately supported C-frame electromagnet of the GMW-5403 class supplies a reference static field of approximately 0.196 T at approximately 20 A and an 80-mm gap; permanent-magnet/yoke and other mapped field sources are alternatives. The locked drive chain includes a synthesizer, modulator, amplifier, directional coupler, incident/reflected and, for a two-port structure, transmitted-power detection, and controller. Loaded Q, ring-up, ring-down, absorbed power, and specimen temperature are measured rather than inferred from source power.

E2 — Multi-clock metrology frame (FIGS. 7–8). Local clocks A and C, physically dissimilar (Definition 7), are mounted at the module's registered witness position(s) on a defined local metrology frame coupled to the driven assembly while shielded and isolated from drive leakage, fields, heat, vibration, optical pickup, and electrical coupling to the extent compatible with the measurement. Their different systematic sensitivities are calibrated before drive tests and stored as the witness-clock sensitivity coefficients of the map. Remote clock B is outside the mapped coupling domain. Optical or fiber links and electrically isolated supplies are preferred. The controller records all three pairwise ratios y_{AB} , y_{CB} , y_{AC} continuously, along with clock servo states, link delays, reference-distribution

signals, and environmental measurements at each clock. Clock and assembly positions are exchanged in registered blocks. For the strongest common-response branch, a second independently operated remote clock D or a registered substitution of remote clock and link is also compared, so a B-only frequency shift cannot masquerade as an equal A/B and C/B response. The driven medium's precession output is logged only as a resonance/phase diagnostic.

E3 — Randomized control matrix (FIG. 9). A discovery run explores frequency, field, power, duty, and timing. The analysis model, target contrasts, nuisance regressors, allowed exclusions, and acceptance thresholds are then frozen. Confirmation blocks are randomized and optionally blinded among: drive off; static field only; drive only; resonant drive; detuned drive matched by absorbed energy and temperature; phase-scrambled or sham drive; field-sense reversal; active specimen; substantially nonmagnetic reference; magnetic reference; specimen and clock position swaps; shielding changes; and positive injections of known clock, link, magnetic, thermal, vibration, acceleration, and force artifacts. Multiple-comparison correction and effective independent sample counts are specified before labels are revealed.

E4 — Synchronous comparison and classifier (FIG. 11). The controller estimates drive-correlated contrasts in y_{AB} , y_{CB} , and y_{AC} , fits the nuisance-state vector, and reports one of:

1. null response or upper bound;
2. identified clock, link, magnetic, thermal, mechanical, or electrical artifact;
3. unexplained clock-specific residual;
4. unexplained common fractional residual not yet independently confirmed; or
5. validated candidate common rate response after all applicable gates and replication.

No single-clock shift and no directly driven precession shift is labeled proper time. Only class 5 points enter the map as validated nonzero operating points; class 1 points enter as null points.

E5 — Joint force/rate embodiment (FIG. 10). The driven specimen may be mounted on a sample-force sensor while the cavity, alignment source, feed, and source frame are mechanically supported separately and preferably instrumented by a reaction-force sensor. An outer sensor may measure the aggregate external support. Field magnitude and gradient, vibration, acceleration, strain, temperature, pressure, and drive leakage are logged at the clocks and specimen. Clock/force correlation is evaluated against a causal nuisance model and is corroborative only.

E6 — Validation gates (FIG. 11). A nonzero common response advances to the map only if:

- **G1 — drive verification:** resonance, mode, field, phase, Q, absorbed energy, and timing are measured for each state;
- **G2 — clock integrity:** each clock's systematic-shift and servo budget closes below the registered response threshold;
- **G3 — universality triangle:** the two local-to-remote responses agree while the local ratio remains unchanged within combined uncertainty;

- **G4 — remote-reference and link integrity:** reciprocal or redundant links, a second independent remote reference or registered remote-reference substitution, and link monitors bound remote-clock, distribution, delay, and nonreciprocity artifacts;
- **G5 — environmental equivalence:** matched controls and nuisance fits bound magnetic, thermal, optical, mechanical, pressure, and electrical alternatives;
- **G6 — swaps and reversals:** specimen, clock, reference, position, field-sense, shielding, and selected drive-state swaps behave according to the frozen analysis;
- **G7 — force accounting:** where force is measured, sample, reaction, acceleration, and field-gradient channels do not identify an ordinary cause;
- **G8 — statistical confirmation:** a frozen blinded/held-out analysis survives multiplicity correction and reports uncertainty rather than instrument readability alone;
- **G9 — independent confirmation:** a separately assembled apparatus, preferably operated by an independent laboratory, reproduces the response under a frozen protocol within the registered tolerance; and
- **G10 — spatial validation:** any operating point at a position other than the reference witness position, and any payload-region claim, requires a measured spatial rate-response map with inactive-module, sham, distance, orientation, and shielding controls.

Failure of any applicable gate returns the point to a narrower classification and excludes it from the map as a validated nonzero point. A null remains a valid endpoint and a valid map entry.

E7 — Map storage and conditional controller unlock (FIG. 12). After a validated response, the qualification controller stores R with its uncertainty, sign, gain, bandwidth, hysteresis, saturation, thermal limits, energy account, and recalibration interval as the map of the unit or class, and computes the permissible interpolation region from validated and held-out points. The product controller (620) accepts a setpoint only inside that region, monitors continuing gate conditions through the runtime witnesses, and returns to safe-off if a limit fails. A proper-time label is unavailable on these measurements alone: candidate-common and spatial gates are necessary but not sufficient, and a supported interpretation requires independent non-clock or process-rate universality evidence (Definition 34).

E8 — Spatial and array qualification (FIG. 13). One or more modules surround or adjoin the intended payload region. Dissimilar clock pairs are placed at registered locations inside, outside, and near the module array, with reciprocal reference links and inactive, sham, shielded, sequential, in-phase, and phase-scrambled module states. The apparatus maps response versus position, distance, orientation, shielding, module count, and module phase without presuming additivity or an enclosure effect, producing the spatial rate-response map and array allocation map for G4 and G6. Only a replicated map showing a common response throughout a payload region enables the G4 setpoint mode for that region. Perishable, chemical, biological, and other process-rate uses remain prophetic until a process-rate map (G5) exists; no slower-aging product is represented as constructed or enabled by current data.

E9 — Redundant comparator network (FIG. 14). At least four clock/reference nodes with sufficient links, or three nodes with independently realized parallel links, form a comparison graph whose cycle-space dimension is at least two. The controller computes independent cycle residuals in log-ratio space, synchronizes the measurements, and compares the residual vector with a registered incidence/sensitivity matrix. One loop detects inconsistency but does not localize it. Fault localization is reported only when the network supplies sufficient independent cycles and the candidate fault columns are distinguishable at the measured uncertainty. In deployment, the same test runs on the network formed by the witness clock(s), reference, and any second reference or redundant link present at the comparison port.

E10 — Materials screening. YIG is the reference species. Substituted garnets and other ordered media are candidates only after composition, dimensions, alignment state, resonance spectrum, linewidth, coherence, dielectric loss, magnetostriction, thermal response, and clock systematics are measured under E1–E6 and the resulting unit or class is qualified on its own map. Meeting a coherence threshold alone does not establish a rate response.

PROPHETIC EXAMPLES (paper examples; no experiment, nonzero response, qualified unit, or controlled offset is represented as achieved)

The following are paper examples. No apparatus described below is represented as having produced a nonzero common rate or proper-time response, and no unit is represented as having been qualified. Every number follows from the stated noise model and integration time.

P1 — Qualification commissioning sensitivity. Two independent clocks each having idealized white fractional instability $1 \times 10^{-11}/\sqrt{\tau}$ are compared in equal ON/OFF allocation. For total elapsed time T , the idealized ON-minus-OFF uncertainty is approximately $2\sqrt{2} \times 10^{-11}/\sqrt{T}$. At $T = 10^5$ s, that value is approximately 8.9×10^{-14} , so a 10^{-13} contrast is only about 1.1 standard deviations. An idealized three-standard-deviation observation of that contrast requires at least 7.2×10^5 s (about 8.3 days), before drift, dead time, multiplicity, and systematic uncertainty. Accordingly, 10^{-13} is an instrument-design target, not a predicted response or promised commissioning result. The actual run length follows the measured Allan deviation and full uncertainty budget.

P2 — Artifact challenge. Known magnetic, thermal, strain, acceleration, reference-link, and servo perturbations are injected separately and in combinations. The classifier must identify their clock-specific or common-mode signatures before any unexplained residual is evaluated. Resonant and detuned/phase-scrambled states are matched by absorbed energy and temperature history rather than source power alone. The specimen, clocks, and link endpoints are swapped in blinded blocks.

P3 — G1 module in service (conditional). Suppose qualification of a unit returns one validated nonzero operating point y_1 with uncertainty u_1 at the witness bay, replicated under G9, and null points elsewhere. The stored permissible region is then the single point y_1 (and the drive-off baseline); the operator may command $y_{\text{set}} \in \{0, y_1\}$ only. With a deployed witness clock of the same idealized $1 \times 10^{-11}/\sqrt{\tau}$ instability against a reference of equal or better instability, resolving y_W to a one-standard-deviation uncertainty of 10^{-13} requires an averaging interval $\tau = (10^{-11}/10^{-13})^2 = 10^4$ s; the regulation bandwidth of step (f) is

therefore bounded at about 10^{-4} Hz for that witness, and the report at shorter intervals carries the correspondingly larger uncertainty. The value of y_1 , its sign, and whether any such point exists are qualification outputs not predicted here.

P4 — G2 transportable source (conditional). A unit qualified as in P3 is transported; post-transport ratio checks at y_1 against its internal witness must agree with the map within the stored tolerance before the unit presents a step to a receiving comparator. The receiving comparator's indicated step is compared with the commanded $\Delta y = y_1 \pm u_1$; a disagreement beyond the combined uncertainty is a finding about the receiving comparator or link only after the source's own runtime artifact test passes. No set of commanded offsets, drive power, sign, or scaling law is predicted here.

P5 — G3 physical-clock steering (conditional). A customer hyperfine standard whose sensitivity coefficients are recorded in the map is mounted at the witness position and steered to zero rate offset against a remote reference. Where the required correction lies within $[-|y_1|, |y_1|]$ and the map contains points of the required sign, the steering controller commands the module; otherwise the electronic path carries the correction. The report states the physical and electronic portions separately. On a comparison-link fault the module safe-offs and the electronic path assumes the full correction within one control cycle; the clock output is uninterrupted.

P6 — G4 payload-region enclosure qualification (conditional). Six modules are arranged about a test region with witness clock pairs at the center, boundary, and exterior and remote references outside the mapped coupling domain. Sequential, simultaneous, phase-scrambled, inactive, sham, distance, and shielding states determine whether any response is local to the witness hardware or follows a reproducible spatial law. Only positions at which G10 passes define the payload region; the array allocation map records whether the six-module response exceeds, equals, or falls short of any single-module response—additivity is a result. Payload aging, chemistry, or biology is not inferred from the interior witnesses; those are G5 processes requiring their own process-rate map.

P7 — G6 paired differential source (conditional). Two qualified modules whose maps each contain a validated point of the same magnitude and opposite sign are mounted on a common frame with measured cross-coupling. Commanding $(+y_1, -y_1)$ presents a differential step $2y_1$ to a direct witness-to-witness comparator while each witness-to-reference channel reports $\pm y_1$; the direct differential channel rejects reference drift common to both. If the maps contain only one sign, the commanded pair is $(y_1, 0)$ and the differential is y_1 . Neither the existence of opposite-sign points nor their magnitude is predicted.

POAMS-NATIVE DIRECT RATE-CONTROL EMBODIMENT

The universality triangle and empirical rate-response maps are powerful qualification species, not definitions of time. Under POAMS, time is the rate of angular change itself. A resonantly driven ordered population may therefore be operated prophetically to reallocate internal rate into coherent winding and command the local rate state directly. Extant proper-time criteria, metric language, or agreement among conventional clock technologies are not prerequisites of that native mechanism, although multiple clocks remain useful controls. No fixed magnitude or successful result is represented as achieved.

CLAIMS

1. A rate-control module comprising: a body of ordered medium; an alignment-field source configured to apply to the body a static alignment field of commanded magnitude and sense; a coupling structure and drive source configured to apply to the body an oscillating drive at a located precession resonance of the body's aligned internal-angular-momentum population; a fixture defining a registered witness position within a mapped response region of the module and outside the coupling structure; a witness clock at the registered witness position; a comparison port configured to compare an output of the witness clock with a reference outside a mapped coupling domain of the module and to record a link status; nuisance sensors comprising at least field and field-gradient, drive-leakage, and temperature sensors at the registered witness position and drive-power detectors; a calibration memory storing an empirical rate-response map relating a drive state of the module to a fractional frequency-ratio response of a clock at the registered witness position relative to the reference, the map comprising at least one independently validated nonzero operating point with an uncertainty, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters; a controller configured to accept a fractional rate setpoint, to inhibit the setpoint when it or a current environmental state lies outside the permissible interpolation region, to determine from the map a drive state for the setpoint, to apply the oscillating drive at the determined drive state, to regulate on a measured witness-to-reference fractional response, to compare runtime witnesses comprising outputs of the nuisance sensors and the link status with the runtime limits, to report an achieved witness-to-reference fractional response with propagated uncertainty and a validity state, and to transition the module to a safe-off state in which the oscillating drive is off upon a runtime-limit exceedance, a link failure, or a fault; and a hardware drive interrupt reachable from every state of the controller.
2. The module of claim 1, wherein the controller is further configured to compute, from sensitivity coefficients of the witness clock stored in the calibration memory and from the measured field, field gradient, drive leakage, temperature, and servo state at the registered witness position, a predicted ordinary perturbation of the witness clock, and to report an invalid state when the measured witness-to-reference fractional response is accounted for by the predicted ordinary perturbation within the stored uncertainty.

3. The module of claim 1, further comprising a second witness clock of a technology physically dissimilar to the witness clock at a second registered witness position, the controller being configured to report a runtime universality check comprising the difference between the two witness-to-reference fractional responses with uncertainty.
4. The module of claim 1, wherein the controller is configured to interleave, at a stored interval, drive-off and detuned absorbed-energy-matched zero checks with setpoint operation, to transition to a degraded state and shrink the permissible interpolation region when a zero check drifts beyond a stored tolerance, and to set a recalibration flag.
5. A method of operating a rate-control module having a body of ordered medium, an alignment-field source, a coupling structure and drive source, a registered witness position, a witness clock at the registered witness position, a comparison port to a reference outside a mapped coupling domain, nuisance sensors, and a calibration memory storing an empirical rate-response map having at least one independently validated nonzero operating point, an uncertainty, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters, the method comprising: loading the map and verifying that the provenance record matches an identifier of the module and that a technology and position of the witness clock match the map; executing a drive-off zero check of the witness-to-reference fractional response against a stored baseline; receiving a fractional rate setpoint; confirming that the setpoint and a current environmental state lie within the permissible interpolation region and otherwise inhibiting the setpoint; determining from the map a drive state for the setpoint; applying an oscillating drive to the body at a located precession resonance at the determined drive state within stored slew and bandwidth limits; regulating on the measured witness-to-reference fractional response; comparing runtime witnesses comprising outputs of the nuisance sensors and a link status of the comparison port with the runtime limits on each control cycle; reporting an achieved witness-to-reference fractional response with propagated uncertainty and a validity state; and transitioning to a safe-off state in which the oscillating drive is off upon a runtime-limit exceedance, a link failure, or a fault, wherein the method is performed without executing a qualification protocol that established the map.
6. The method of claim 5, further comprising bounding a regulation bandwidth of the regulating at a bandwidth at which the witness-to-reference fractional response is resolved, from the measured instability of the witness clock and reference, to a stored tolerance.
7. The method of claim 5, further comprising performing a service recalibration comprising a drive-off zero check, a detuned zero check, a witness-to-reference ratio check at one or more validated operating points of the map, and a comparison-link reciprocity check, and disabling the setpoint until the module is returned to full qualification when any check exceeds a stored tolerance.
8. A transportable rate-step calibration source comprising: a transport case; within the transport case, a rate-control module having a body of ordered medium, an alignment-field source configured to apply a static alignment field of commanded magnitude and sense, a coupling structure and drive source configured to apply an oscillating drive at a located precession resonance of the body's aligned internal-angular-momentum population, a fixture defining a first registered witness

position, a witness clock at the first registered witness position, a comparison port to a reference outside a mapped coupling domain of the module, and nuisance sensors at the first registered witness position; a receiving fixture defining a second registered witness position for a receiving clock, or a comparison-port adapter for a receiving comparator; transport sensors comprising at least shock, temperature, and field-exposure sensors and a transport record; a calibration memory storing an empirical rate-response map relating a drive state of the module to a fractional frequency-ratio response of a clock at the first and at the second registered witness positions, the map comprising at least one independently validated nonzero operating point at the second registered witness position with an uncertainty, a provenance record, a permissible interpolation region, runtime limits, safe-off parameters, and a stored transport limit; and a controller configured to step the module between a drive-off baseline and a validated operating point of the map so as to present to the receiving clock or receiving comparator a commanded fractional rate step of stated value and uncertainty without an electronic frequency offset in a signal path of the receiving clock, to compare the witness-to-reference fractional response of the witness clock with the commanded step in real time and invalidate the presented step upon a deviation beyond a stored tolerance, and to report an invalid state when the transport record exceeds the stored transport limit until a post-transport ratio check at a validated operating point agrees with the map.

9. The calibration source of claim 8, wherein the alignment-field source is a permanent-magnet yoke with a trim winding, and the controller is configured to step the module in a sequence of a plurality of validated operating points and dwell intervals stored as a profile.
10. The calibration source of claim 8, wherein the provenance record traces the commanded fractional rate step to remote reference clocks used at qualification and the report identifies the step as a commanded fractional rate step at the second registered witness position.
11. A method of checking a frequency comparator or a receiving clock with a transportable rate-step calibration source having a rate-control module with a body of ordered medium, an alignment-field source, a coupling structure and drive source, a witness clock at a first registered witness position, a comparison port, a receiving fixture defining a second registered witness position, transport sensors, and a calibration memory storing an empirical rate-response map having at least one independently validated nonzero operating point at the second registered witness position, the method comprising: transporting the source; verifying that a transport record of the transport sensors lies within a stored transport limit; performing a post-transport ratio check of the witness clock against a reference at the validated operating point and confirming agreement with the map within a stored tolerance; placing a receiving clock at the second registered witness position or connecting a receiving comparator through the comparison port; stepping the module between a drive-off baseline and the validated operating point while recording the witness-to-reference fractional response and an indicated step of the receiving clock or receiving comparator, without applying an electronic frequency offset in a signal path of the receiving clock; comparing the indicated step with the commanded fractional rate step and its uncertainty from the map; and reporting a comparator gain, step response, or link consistency finding only when the witness-to-

reference fractional response agreed with the commanded step within the stored tolerance during the stepping.

12. The method of claim 11, further comprising repeating the stepping at a plurality of dwell intervals and reporting a settling time of the receiving comparator.
13. A physical-clock steering system comprising: a rate-control module having a body of ordered medium, an alignment-field source configured to apply a static alignment field of commanded magnitude and sense, a coupling structure and drive source configured to apply an oscillating drive at a located precession resonance of the body's aligned internal-angular-momentum population, a fixture defining a registered witness position, and nuisance sensors at the registered witness position; a steered clock at the registered witness position, the steered clock having sensitivity coefficients recorded in a calibration memory; a reference input receiving a reference from outside a mapped coupling domain of the module; a comparator measuring a fractional frequency ratio of the steered clock to the reference; the calibration memory storing an empirical rate-response map relating a drive state of the module to a fractional frequency-ratio response of a clock at the registered witness position, the map comprising at least one independently validated nonzero operating point with an uncertainty, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters; an electronic steering path configured to apply a frequency correction to an output of the steered clock; and a steering controller configured to compute a correction required to hold the measured fractional frequency ratio at a commanded value, to partition the correction into a module portion lying within the permissible interpolation region and an electronic portion, to command the module at a drive state determined from the map for the module portion and the electronic steering path for the electronic portion, to compare runtime witnesses comprising outputs of the nuisance sensors with the runtime limits, to report the module portion and the electronic portion separately with uncertainty, and, upon a runtime-limit exceedance, a loss of the reference, or a fault, to transition the module to a safe-off state in which the oscillating drive is off and to transfer the entire correction to the electronic steering path or to a hold-over state without interrupting the output of the steered clock.
14. The steering system of claim 13, wherein the steering controller is configured to compute, from the recorded sensitivity coefficients and measured field, field gradient, drive leakage, temperature, and servo state at the registered witness position, a predicted ordinary perturbation of the steered clock, and to attribute to the module portion only that part of the measured change in the fractional frequency ratio not accounted for by the predicted ordinary perturbation.
15. The steering system of claim 13, further comprising a witness clock of a technology physically dissimilar to the steered clock at a second registered witness position, the steering controller being configured to report a runtime universality check comprising the difference between the steered-clock and witness-clock fractional responses to the reference.
16. A method of steering a clock, comprising: mounting a steered clock having recorded sensitivity coefficients at a registered witness position of a rate-control module having a body of ordered medium, an alignment-field source, a coupling structure and drive source, and nuisance sensors,

the module having a calibration memory storing an empirical rate-response map with at least one independently validated nonzero operating point, a permissible interpolation region, runtime limits, and safe-off parameters; measuring a fractional frequency ratio of the steered clock to a reference from outside a mapped coupling domain of the module; computing a correction required to hold the measured fractional frequency ratio at a commanded value; partitioning the correction into a module portion lying within the permissible interpolation region and an electronic portion; applying an oscillating drive to the body at a located precession resonance at a drive state determined from the map for the module portion; applying the electronic portion as a frequency correction to an output of the steered clock; comparing runtime witnesses comprising outputs of the nuisance sensors with the runtime limits on each control cycle; reporting the module portion and the electronic portion separately with uncertainty; and, upon a runtime-limit exceedance, a loss of the reference, or a fault, turning the oscillating drive off and transferring the entire correction to the electronic portion or to a hold-over state without interrupting the output of the steered clock.

17. The method of claim 16, wherein the commanded value is zero and the reporting identifies a physically steered fractional rate relative to the reference without asserting a change of proper time of the steered clock.
18. A payload-region rate-control enclosure comprising: a plurality of rate-control modules, each having a body of ordered medium, an alignment-field source configured to apply a static alignment field of commanded magnitude and sense, and a coupling structure and drive source configured to apply an oscillating drive at a located precession resonance of the body's aligned internal-angular-momentum population, the modules being arranged on a common frame about a defined payload region; a plurality of fixtures defining registered witness positions at interior, boundary, and exterior locations relative to the payload region; witness clocks at the registered witness positions; a comparison port configured to compare each witness clock with a common reference outside a mapped coupling domain of the enclosure; nuisance sensors at the registered witness positions; a calibration memory storing a spatial rate-response map relating per-module drive states to a fractional frequency-ratio response at each registered witness position with uncertainty, the spatial map having been established with inactive-module, sham-module, sequential, in-phase, phase-scrambled, distance, orientation, and shielding qualification states and comprising at least one independently validated nonzero operating point at each interior registered witness position, an array allocation map with cross-talk terms, a provenance record, a permissible interpolation region, runtime limits, an exterior-witness limit, and safe-off parameters; and an array controller configured to accept a common fractional rate setpoint for the payload region, to inhibit the setpoint when it or a current environmental state lies outside the permissible interpolation region, to determine per-module drive states from the array allocation map, to apply the oscillating drives at the determined drive states, to regulate on the measured fractional responses of the interior witness clocks, to report the achieved fractional response at each registered witness position with uncertainty and a spatial uniformity residual, and to transition all modules to a safe-off state in which the oscillating drives are off upon a disagreement among the interior witness clocks beyond

a stored tolerance, an exterior witness response beyond the exterior-witness limit, a runtime-limit exceedance, or a fault.

19. The enclosure of claim 18, wherein the payload region is defined in the calibration memory as the set of registered witness positions at which the spatial map contains an independently validated nonzero operating point, and the array controller inhibits any report of a fractional response at a position outside that set.
20. The enclosure of claim 18, wherein at least two witness clocks at interior registered witness positions are of physically dissimilar technologies and the array controller reports a runtime universality check comprising their difference with uncertainty.
21. A method of operating a payload-region rate-control enclosure having a plurality of rate-control modules arranged about a defined payload region, witness clocks at registered interior, boundary, and exterior witness positions, a comparison port to a common reference, nuisance sensors, and a calibration memory storing a spatial rate-response map and an array allocation map having at least one independently validated nonzero operating point at each interior registered witness position, a permissible interpolation region, runtime limits, an exterior-witness limit, and safe-off parameters, the method comprising: loading the spatial map and the array allocation map and verifying a provenance record against an identifier of the enclosure; executing a drive-off zero check of each witness clock against the common reference; receiving a common fractional rate setpoint for the payload region; confirming that the setpoint and a current environmental state lie within the permissible interpolation region and otherwise inhibiting the setpoint; determining per-module drive states from the array allocation map; applying an oscillating drive at a located precession resonance in each module at its determined drive state; regulating on the measured fractional responses of the interior witness clocks; comparing runtime witnesses comprising outputs of the nuisance sensors, the boundary and exterior witness responses, and a link status with the runtime limits and the exterior-witness limit; reporting the achieved fractional response at each registered witness position with uncertainty and a spatial uniformity residual; and turning all oscillating drives off upon a disagreement among the interior witness clocks beyond a stored tolerance, an exterior witness response beyond the exterior-witness limit, a runtime-limit exceedance, or a fault.
22. The method of claim 21, wherein the reporting identifies the achieved fractional response as a witness-clock response at registered positions and withholds any statement regarding a rate of a process within the payload region.
23. A process-rate system comprising: a rate-control module having a body of ordered medium, an alignment-field source configured to apply a static alignment field of commanded magnitude and sense, a coupling structure and drive source configured to apply an oscillating drive at a located precession resonance of the body's aligned internal-angular-momentum population, a fixture defining a registered witness position, and nuisance sensors at the registered witness position; a witness clock at the registered witness position; a comparison port to a reference outside a mapped coupling domain of the module; a holder positioning a defined non-clock process at the registered witness position; a process-rate sensor configured to measure a rate index of the defined process

independently of the witness clock; environmental sensors of the defined process; a calibration memory storing an empirical rate-response map of the witness clock and a process-rate map relating a drive state of the module to the rate index of the defined process, the process-rate map comprising at least one independently validated nonzero operating point established with the defined process at the registered witness position under randomized drive and control states and independently of the rate-response map of the witness clock, an uncertainty, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters; and a controller configured to accept a process-rate setpoint, to inhibit the setpoint when it or a current environmental state lies outside the permissible interpolation region, to determine from the process-rate map a drive state, to apply the oscillating drive at the determined drive state, to regulate on the measured witness-to-reference fractional response, to compare runtime witnesses comprising outputs of the nuisance sensors, the environmental sensors of the defined process, and a link status with the runtime limits, to report an achieved rate-index ratio of the defined process from the process-rate sensor with uncertainty together with the witness-to-reference fractional response, and to transition the module to a safe-off state in which the oscillating drive is off upon a runtime-limit exceedance or a fault.

24. The process-rate system of claim 23, wherein the controller is configured to report an artifact state, and to withhold the achieved rate-index ratio, when a change in the rate index is accounted for by a change in an output of the environmental sensors of the defined process within a stored process-sensitivity model.
25. A method of operating a process-rate system having a rate-control module with a body of ordered medium, an alignment-field source, a coupling structure and drive source, a registered witness position, a witness clock and a defined non-clock process at the registered witness position, a process-rate sensor, environmental sensors of the defined process, a comparison port to a reference, nuisance sensors, and a calibration memory storing a process-rate map having at least one independently validated nonzero operating point for the defined process, a permissible interpolation region, runtime limits, and safe-off parameters, the method comprising: loading the process-rate map and verifying a provenance record against identifiers of the module and of the defined process; executing a drive-off baseline measurement of the rate index of the defined process and of the witness-to-reference fractional response; receiving a process-rate setpoint; confirming that the setpoint and a current environmental state lie within the permissible interpolation region and otherwise inhibiting the setpoint; determining from the process-rate map a drive state; applying an oscillating drive to the body at a located precession resonance at the determined drive state; regulating on the measured witness-to-reference fractional response; comparing runtime witnesses comprising outputs of the nuisance sensors and the environmental sensors of the defined process with the runtime limits; reporting an achieved rate-index ratio of the defined process measured by the process-rate sensor with uncertainty together with the witness-to-reference fractional response; and turning the oscillating drive off upon a runtime-limit exceedance or a fault.

26. The method of claim 25, wherein the reporting is confined to the defined process at the registered witness position and withholds any statement regarding another process or a proper time.
27. A paired differential-rate system comprising: a first rate-control module and a second rate-control module, each having a body of ordered medium, an alignment-field source configured to apply a static alignment field of commanded magnitude and sense, a coupling structure and drive source configured to apply an oscillating drive at a located precession resonance of the body's aligned internal-angular-momentum population, a fixture defining a registered witness position, and nuisance sensors at the registered witness position, the modules being mounted on a common frame at a measured separation; a first witness clock at the registered witness position of the first module and a second witness clock at the registered witness position of the second module; a comparison port to a common reference outside a mapped coupling domain of both modules; a differential comparator configured to measure a fractional frequency ratio of the first witness clock to the second witness clock directly; a calibration memory storing, for each module, an empirical rate-response map comprising at least one independently validated nonzero operating point with uncertainty, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters, and further storing a measured cross-coupling of each module on the witness clock of the other module; and a differential controller configured to accept a first fractional rate setpoint for the first module and a second fractional rate setpoint for the second module differing from the first, to inhibit either setpoint whose value or sign lacks a validated operating point in the map of its module or lies outside its permissible interpolation region, to determine drive states from the maps corrected for the measured cross-coupling, to apply the oscillating drives, to regulate on the measured fractional responses of the witness clocks to the common reference, to report the achieved differential fractional response from the differential comparator with uncertainty together with each witness-to-reference fractional response, to compare runtime witnesses with the runtime limits, and to transition both modules to a safe-off state in which the oscillating drives are off upon a runtime-limit exceedance or a fault in either module.
28. The paired system of claim 27, wherein the differential controller is configured to command setpoints of opposite sign only when the map of each module contains an independently validated operating point of the commanded sign, and otherwise to command a setpoint of zero for the module lacking such a point.
29. A method of operating a paired differential-rate system having first and second rate-control modules each with a body of ordered medium, an alignment-field source, a coupling structure and drive source, a registered witness position, a witness clock, and nuisance sensors, a common reference, a differential comparator between the witness clocks, and a calibration memory storing for each module an empirical rate-response map having at least one independently validated nonzero operating point, a permissible interpolation region, runtime limits, and safe-off parameters, and a measured cross-coupling between the modules, the method comprising: loading the maps and the cross-coupling and verifying provenance records against identifiers of the modules; executing a drive-off zero check of each witness clock against the common reference and

of the differential comparator; receiving a first fractional rate setpoint and a differing second fractional rate setpoint; inhibiting either setpoint whose value or sign lacks a validated operating point in the map of its module or lies outside its permissible interpolation region; determining drive states from the maps corrected for the cross-coupling; applying an oscillating drive at a located precession resonance in each module at its determined drive state; regulating on the measured fractional responses of the witness clocks to the common reference; comparing runtime witnesses with the runtime limits; reporting the achieved differential fractional response measured by the differential comparator with uncertainty together with each witness-to-reference fractional response; and turning both oscillating drives off upon a runtime-limit exceedance or a fault in either module.

30. The method of claim 29, further comprising presenting the achieved differential fractional response to a frequency comparator or link under test as a commanded differential step without an electronic frequency offset, and reporting a comparator or link finding only when the differential comparator agreed with the commanded differential step within a stored tolerance.
31. A method of qualifying a rate-control module having a body of ordered medium, an alignment-field source, a coupling structure and drive source, and a registered witness position, to produce an empirical rate-response map, the method comprising: using the alignment-field source and drive source to establish an alignment state of an internal angular-momentum population of the body, locate a precession resonance, and apply an oscillating drive in randomized drive and control states; operating a first local clock and a physically dissimilar second local clock at the registered witness position in a defined local metrology frame associated with the body; operating a remote reference clock outside a mapped coupling domain of the module; recording the two local-to-remote frequency ratios and the local-to-local frequency ratio together with measured nuisance channels; classifying each drive state as a null response, an identified artifact, a clock-specific residual, or an unexplained common fractional residual under predetermined gates; withholding designation of a candidate common rate response unless the two local-to-remote ratios exhibit equal fractional changes within combined uncertainty while the local-to-local ratio remains unchanged within its uncertainty; and storing in a calibration memory of the module an empirical rate-response map comprising each drive state classified as a null response as a null operating point and each drive state designated a candidate common rate response and reproduced on a separately assembled apparatus as an independently validated nonzero operating point, together with uncertainty, a provenance record, a permissible interpolation region computed from the validated and held-out points, runtime limits, and safe-off parameters, and recording the module as a qualified rate-responsive module when at least one independently validated nonzero operating point is stored and as a null reference otherwise.
32. The method of claim 31, wherein the randomized states include drive off, static field only, drive only, resonant drive, detuned drive, phase-scrambled or sham drive, field-sense reversal, and at least one active/reference-material substitution, and at least one control is matched to a resonant state by measured absorbed energy and temperature history rather than equal source power alone.

33. The method of claim 31, further comprising swapping positions of the local clocks, swapping the body and a reference specimen, changing shielding, or moving a reference-link endpoint in registered randomized blocks, and operating a second independently operated remote reference clock or a registered substitution of the remote reference clock and its link so that a change confined to the remote reference clock is distinguished from a change common to the local clocks.
34. The method of claim 31, further comprising recording magnetic field and field gradient, drive leakage, incident/reflected/transmitted and absorbed drive power, temperature, blackbody environment, strain, vibration, acceleration, pressure, clock servo state, reference distribution, and link delay or nonreciprocity, and storing sensitivity coefficients of each local clock in the map as witness-clock sensitivity coefficients.
35. The method of claim 31, further comprising separating an exploratory scan from a frozen, blinded or held-out confirmation with multiplicity correction, and applying known clock, link, thermal, magnetic, mechanical, or electrical perturbations as positive controls for the classifying.
36. The method of claim 31, wherein an output of the body's driven precession is used as a drive-state diagnostic and is excluded as a witness of a common rate response, and no operating point is stored as a validated nonzero operating point on the basis of one clock transition.
37. The method of claim 31, further comprising measuring a support force of the body and a reaction force on at least part of the coupling structure or alignment-field source, and treating clock/force correlation as corroborative subject to a measured causal nuisance model.
38. The method of claim 31, further comprising, for a plurality of modules arranged about a payload region, placing dissimilar clock pairs at a plurality of registered positions relative to the modules, operating sequential, simultaneous, inactive-module, sham-module, phase-scrambled, distance, orientation, and shielding states, and storing a spatial rate-response map and an array allocation map with cross-talk terms, or an upper bound, without presuming additivity or an enclosure effect, and defining the payload region as the set of registered positions at which an independently validated nonzero operating point was stored.
39. The method of claim 31, further comprising placing a defined non-clock process at the registered witness position, measuring a rate index of the defined process with a process-rate sensor independently of the local clocks under the randomized drive and control states, and storing a process-rate map only when the rate index follows the drive state under the predetermined gates and is reproduced on a separately assembled apparatus.
40. The method of claim 31, wherein the body is an ordered medium other than a single-crystal yttrium-iron-garnet specimen and is admitted only after its composition, dimensions, alignment state, resonance spectrum, linewidth, coherence, dielectric loss, magnetostriction, thermal response, and clock-systematic couplings are measured.
41. A qualification apparatus for a rate-control module, comprising: a mount for a module having a body of ordered medium, an alignment-field source, a coupling structure, a drive source configured to locate and excite a measured precession resonance, and a registered witness position; first and second physically dissimilar local clocks positioned at the registered witness position in a defined

local metrology frame; a remote reference clock outside a mapped coupling domain of the module; frequency-comparison links configured to obtain the two local-to-remote ratios and the local-to-local ratio; magnetic, thermal, mechanical, drive-power, and link-status sensors; and a qualification controller configured to execute randomized drive and control states, to classify each state as a null response, an identified artifact, a clock-specific residual, or an unexplained common residual, and to write to a calibration memory of the module an empirical rate-response map of null operating points and independently validated nonzero operating points with uncertainty, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters.

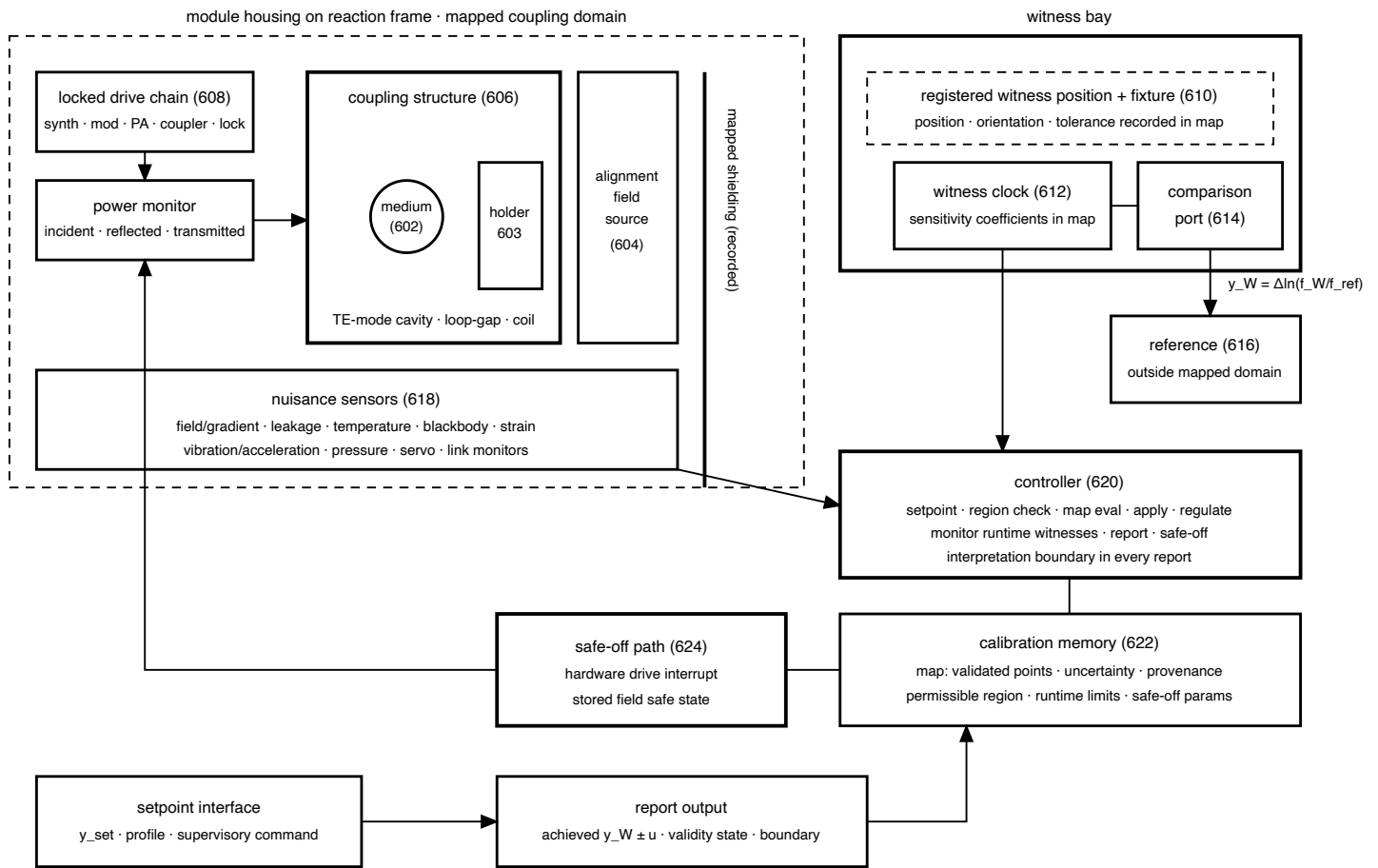
42. The apparatus of claim 41, wherein the body comprises a single-crystal yttrium-iron-garnet sphere, the coupling structure comprises a mapped TE-mode cavity, loop-gap resonator, re-entrant resonator, waveguide, or coil, the alignment-field source comprises a separately supported electromagnet, permanent-magnet yoke, or coil-and-yoke assembly, and the local clocks comprise different atomic transitions or different atomic, crystal, or optical oscillator mechanisms having measured dissimilar field, thermal, strain, optical, and servo sensitivities.
43. The apparatus of claim 41, wherein at least one frequency-comparison link is reciprocal or redundant, a second independently operated remote reference or a registered remote-reference substitution is provided, clock power, control, and data paths are electrically or optically isolated from the coupling structure and drive source, and the apparatus further comprises incident and reflected drive-power detectors and, for a two-port coupling structure, a transmitted-power detector, together with calorimetry or thermal sensing for absorbed-energy matching.
44. The apparatus of claim 41, further comprising a sample-force sensor supporting the body, a reaction-force sensor supporting at least part of the coupling structure, drive source, or alignment-field source, a field-gradient mapping probe, vibration and acceleration sensors, strain sensors, and environmental sensors positioned at the clocks and body, and wherein the qualification controller stores clock sensitivity coefficients, a nuisance-state model, registered classification thresholds, and a frozen confirmation analysis, and reports uncertainty and an upper bound when no response is validated.
45. A method of monitoring a comparator network of a rate-control module or qualification apparatus, comprising: synchronously measuring pairwise frequency ratios in a comparison graph having a cycle-space dimension of at least two, the graph comprising at least four clock or reference nodes or three nodes with independently realized parallel comparison links; computing a plurality of independent cycle residuals in log-ratio space; declaring a comparison inconsistency when a cycle residual exceeds a registered uncertainty; and withholding fault localization unless an incidence or sensitivity matrix has sufficient rank to distinguish the candidate faulted clocks or links.
46. The method of claim 45, wherein clock or link faults are challenged by positive injections and localized from a plurality of independent non-closing cycles rather than one loop, and a declared comparison inconsistency transitions a rate-control module to a degraded or safe-off state.

47. A prophetic POAMS-native rate-control apparatus comprising: a body of ordered medium having an internal angular-momentum population; an alignment-field source; a resonant drive configured to advance a coherent phase of that population; a witness at a registered position; and a controller configured to reallocate internal angular-momentum rate into or out of coherent winding and thereby command the local rate state, without requiring a pre-existing empirical response map or an extant metric transfer law, an achieved rate change or null result being reported by the witness.
48. A method of operating the apparatus of claim 47, comprising: establishing the aligned population; locating and locking a resonance; applying a selected coherent drive state; comparing the witness rate with a drive-off or remote reference; reversing a drive sense, alignment sense, mode, or winding allocation to test the predicted native sign; and reporting the measured local rate change or null result, with multiple dissimilar clocks and a universality triangle used as optional controls rather than definitions of time.

ABSTRACT

Rate-control devices built on a resonantly driven ordered-medium module—an aligned medium, an alignment-field source, a resonance-locked oscillating drive, a registered witness position, a witness clock compared through a comparison port with a reference outside the mapped coupling domain, nuisance sensors, a controller, and a calibration memory holding an empirical rate-response map with independently validated operating points, uncertainty, provenance, permissible interpolation region, runtime limits, and safe-off parameters. Six products and their operating methods: a qualified rate-control module commanding a fractional rate setpoint; a transportable rate-step calibration source; a physical-clock steering system with electronic hand-off; a payload-region enclosure operating from a spatial map; a process-rate system conditional on independent non-clock validation; and a paired differential-rate system. Factory qualification uses two dissimilar local clocks, remote references, a universality triangle, randomized controls, and gates G1–G10; the map admits zero. No proper-time effect is represented as established.

Rate-control module (600) — common product architecture



A unit whose map holds only null points is a null reference: setpoint mode disabled, L0 reported.

FIG. 1

Rate-control module (600): ordered medium, alignment-field source, coupling structure and locked drive chain, registered witness position and witness clock, comparison port to a reference outside the mapped domain, nuisance sensors, controller, calibration memory holding the empirical rate-response map, and hardware safe-off path.

Layered scope L0–L3

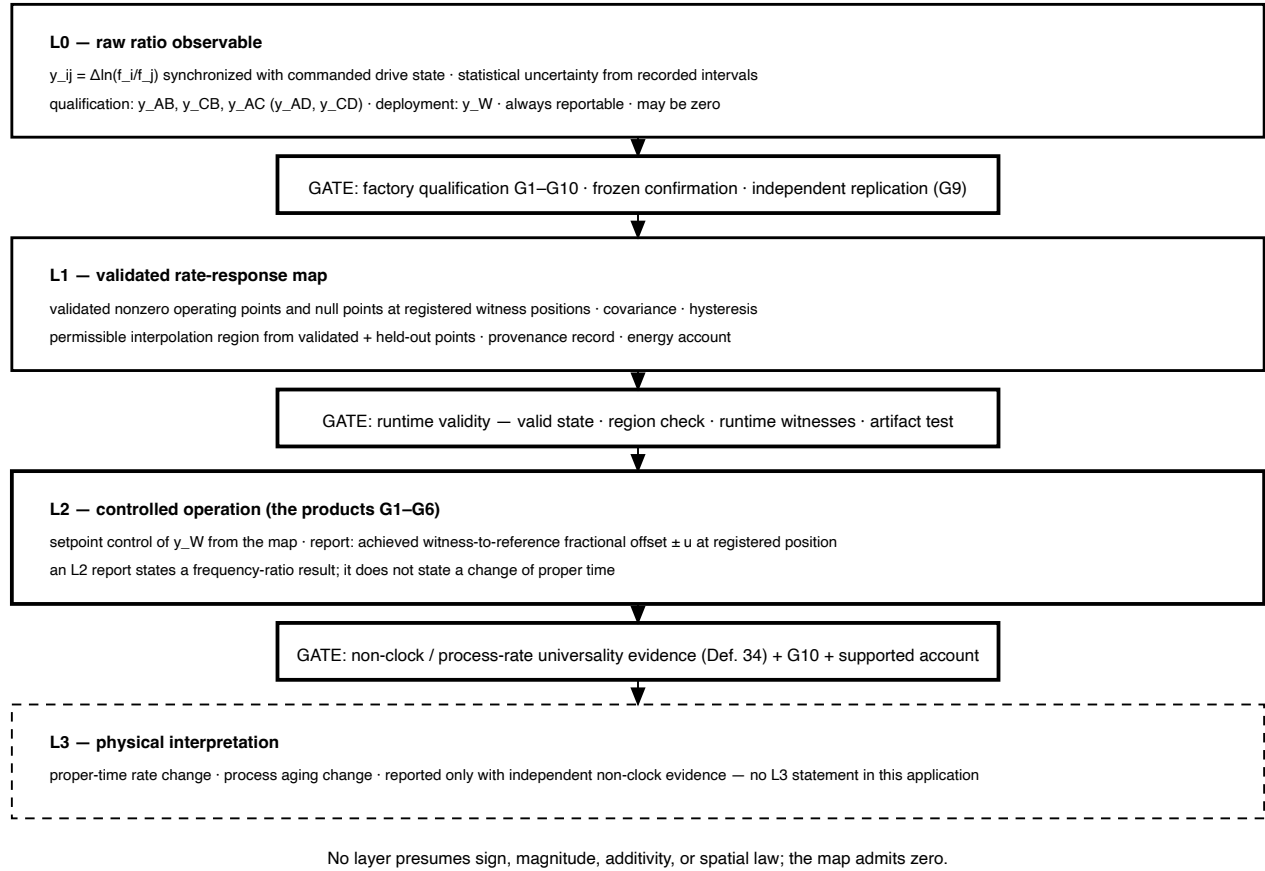
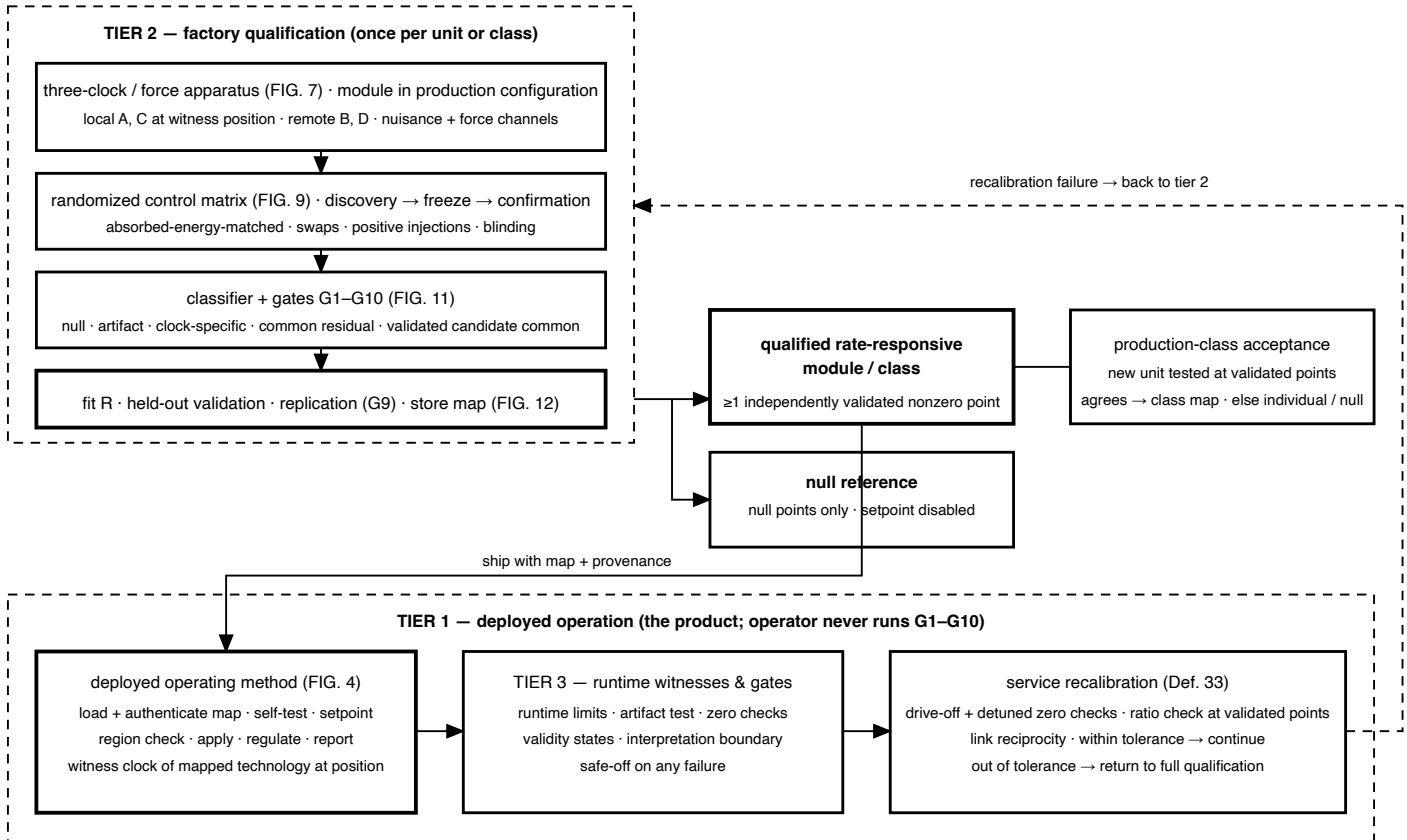


FIG. 2

Layered scope — raw ratio observable, validated map behind the qualification gates, controlled operation behind the runtime validity gate, and physical interpretation behind the non-clock universality gate.

Qualification-to-deployment flow

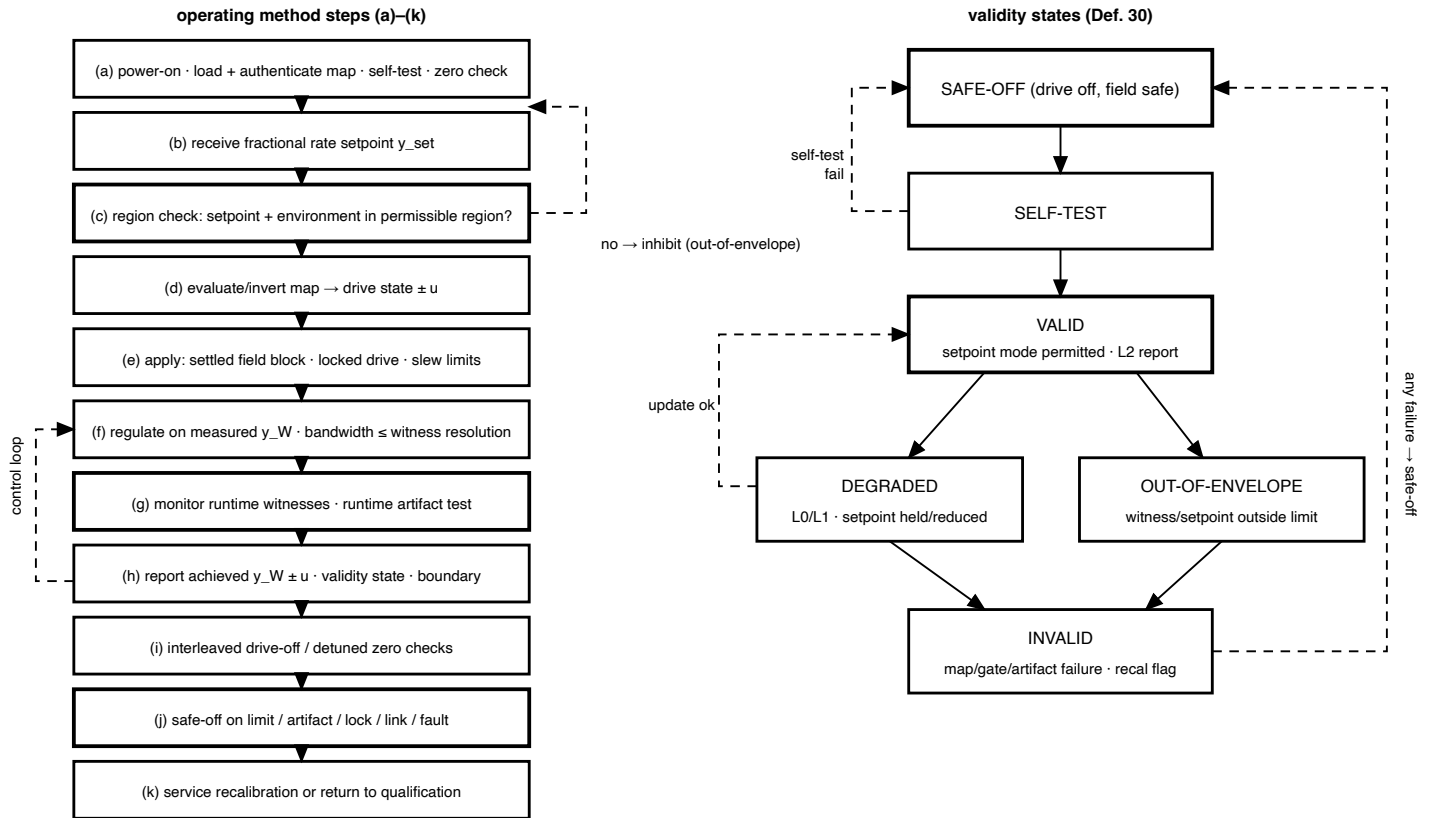


Sign, magnitude, additivity, spatial reach, process effects, and interpretation are outputs of tier 2, never premises of tier 1.

FIG. 3

Qualification-to-deployment flow — factory qualification with the three-clock apparatus produces the map and classifies the unit as a qualified rate-responsive module or a null reference; deployed operation runs from the stored map under runtime witnesses; service recalibration returns a drifted unit to qualification.

Deployed operating method and validity-state machine



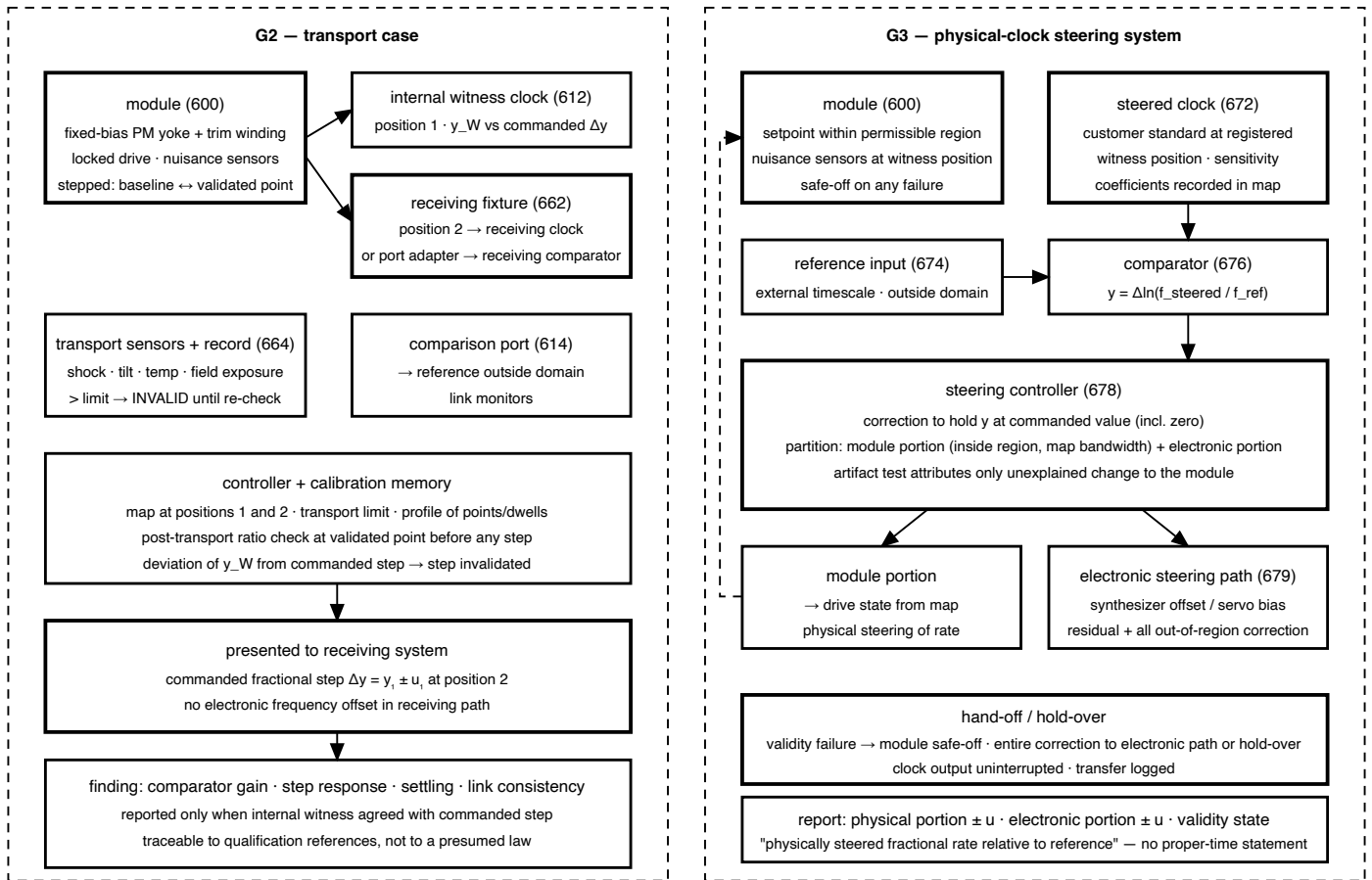
Every report: "achieved witness-to-reference fractional offset at registered position" — the interpretation boundary of Definition 36.

The operator never executes G1–G10; the map is authenticated, not re-derived.

FIG. 4

Deployed operating method steps (a)–(k) and the controller validity-state machine — safe-off, self-test, valid, degraded, out-of-envelope, invalid — with the reporting layer permitted in each state.

G2 transportable rate-step calibration source (660) · G3 physical-clock steering system (670)

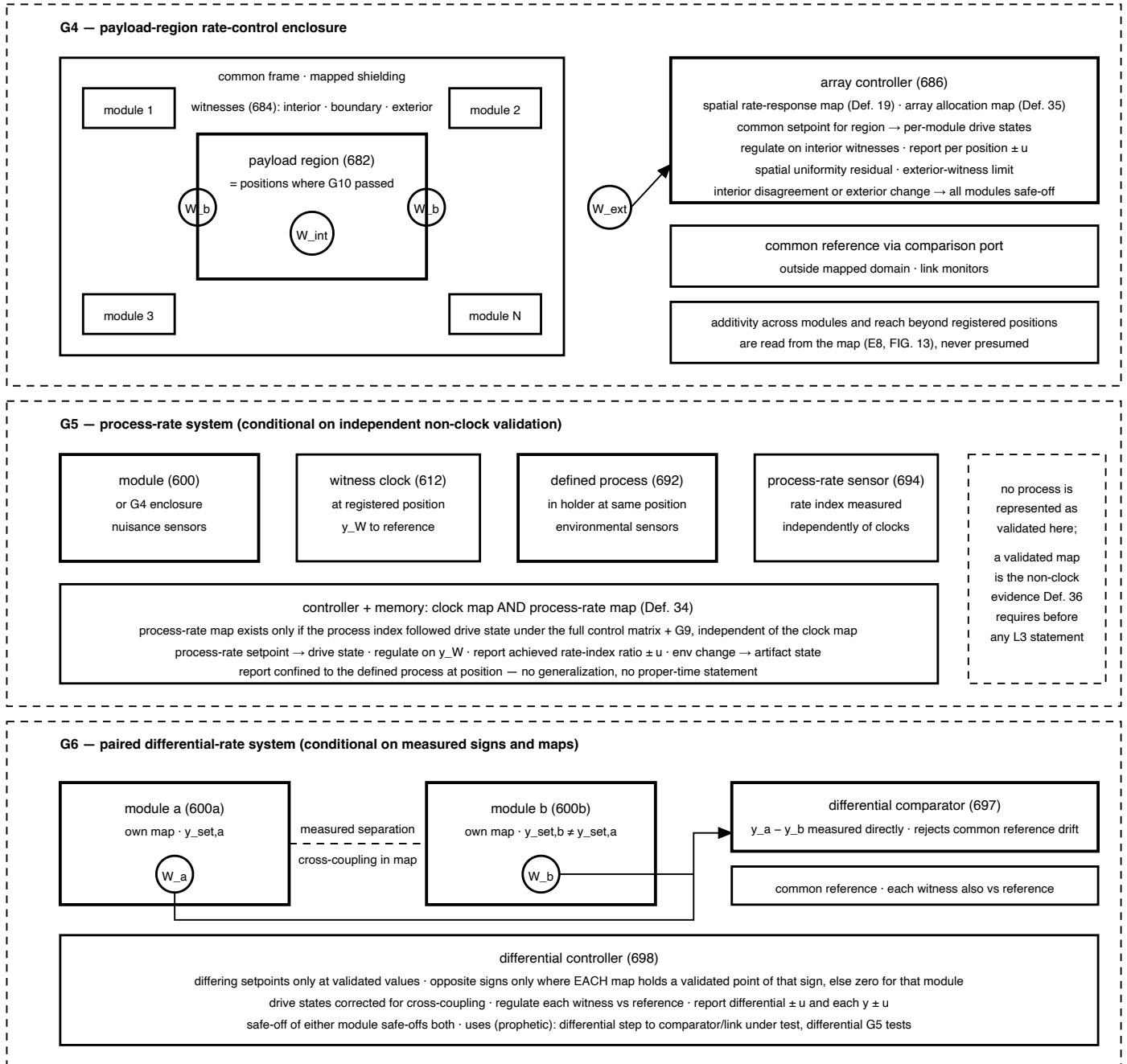


Both products operate only from a stored validated map; a null-qualified unit ships, if at all, as a null reference.

FIG. 5

Left — G2 transportable rate-step calibration source: module with internal witness, receiving fixture or port adapter, transport record, and commanded step presented without an electronic offset. Right — G3 physical-clock steering system: steered clock at the witness position, comparator, steering controller partitioning correction between the module and the electronic path, hand-off and hold-over.

G4 payload-region enclosure (680) · G5 process-rate system (690) · G6 paired differential-rate system (695)

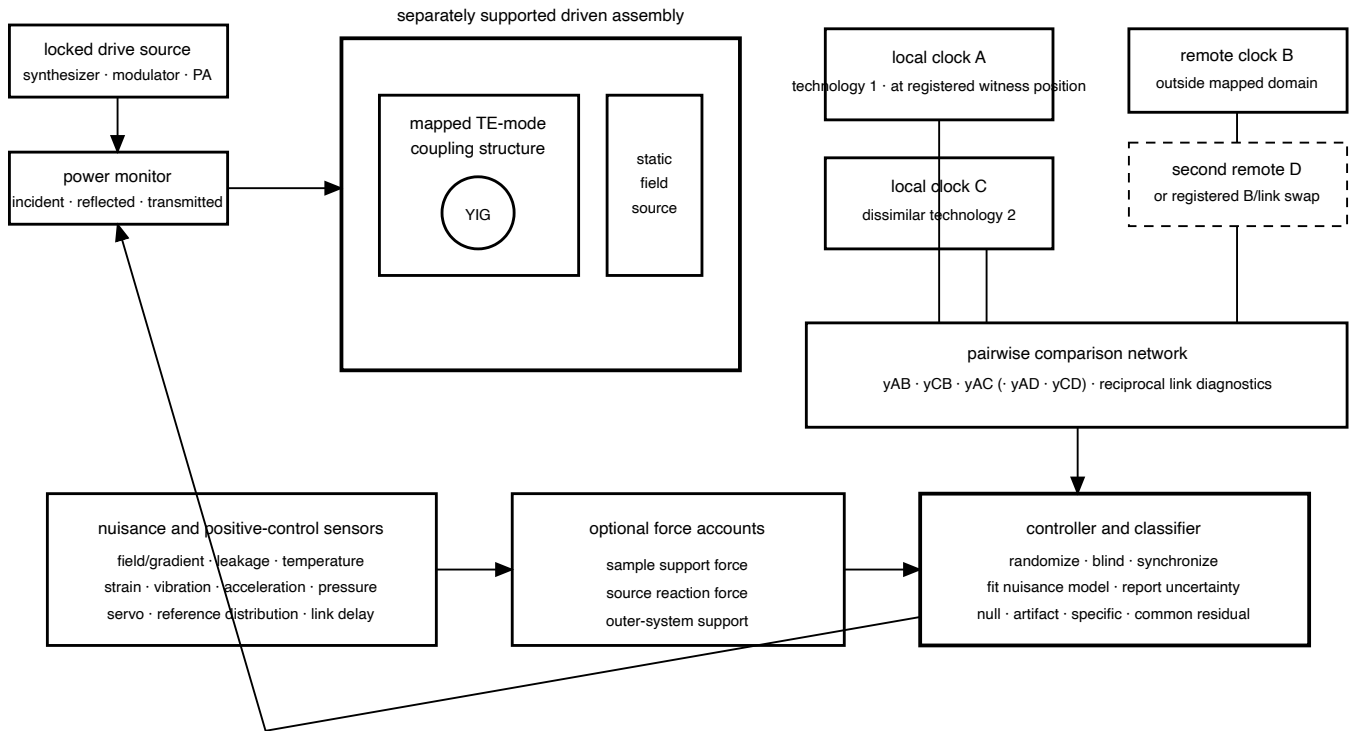


Availability of any sign or magnitude is a qualification result, not a design premise.

FIG. 6

Top — G4 payload-region enclosure: plural modules about a payload region defined by G10, interior/boundary/exterior witnesses, array controller with spatial and allocation maps. Middle — G5 process-rate system: defined process and independent process-rate sensor at the witness position beside the witness clock; process-rate map conditional on independent validation. Bottom — G6 paired differential-rate system: two mapped modules at differing setpoints with a direct differential comparator.

Factory qualification apparatus — three-clock / force characterization of the module

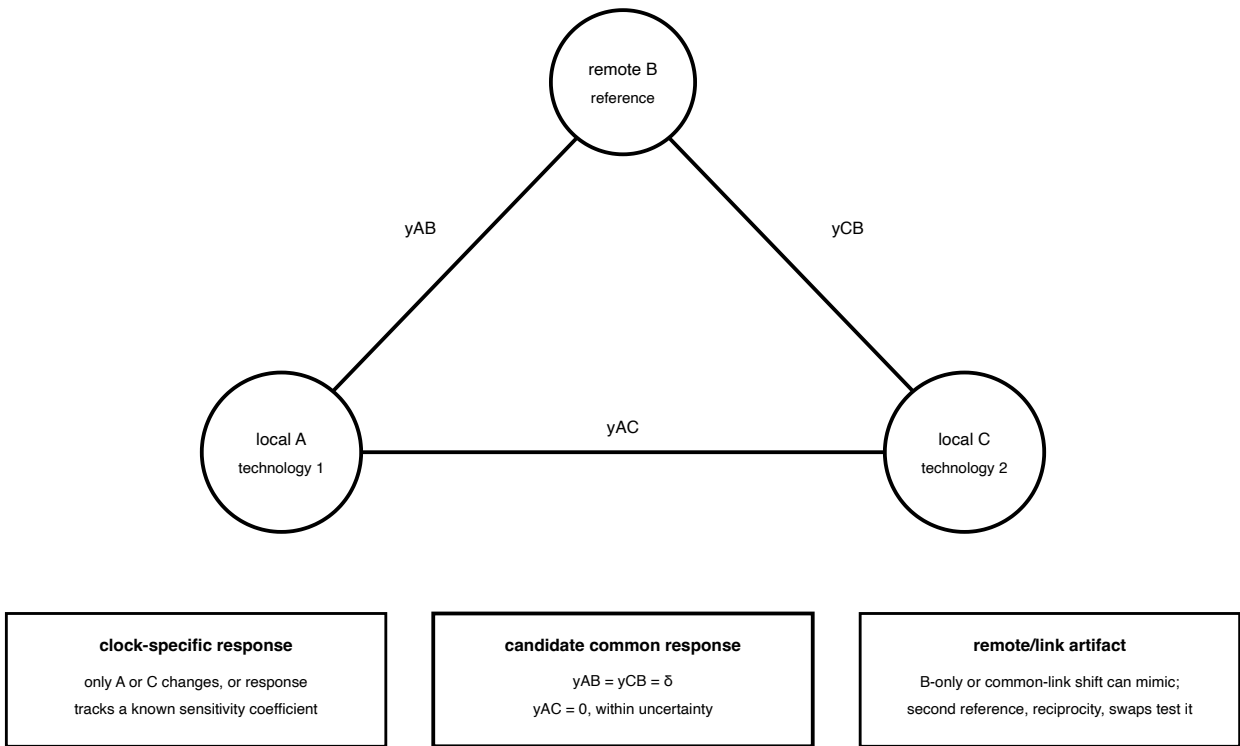


Tier 2: performed once per unit or class; writes the map. Useful when every measured response is zero.

FIG. 7

Factory qualification apparatus: the module under qualification with two dissimilar local clocks at its registered witness position, remote reference B and second remote D (or registered swap), all pairwise ratios, nuisance sensing, optional force accounts, and a zero-valid classifier whose output writes the map.

Universality triangle and competing classifications



The triangle is necessary discrimination—not, by itself, proof of proper time.

FIG. 8

Universality triangle: a candidate common response changes both local-to-remote ratios equally while leaving the local-to-local ratio unchanged; alternative clock and link explanations remain tested.

Randomized control timing and zero-permitted extraction

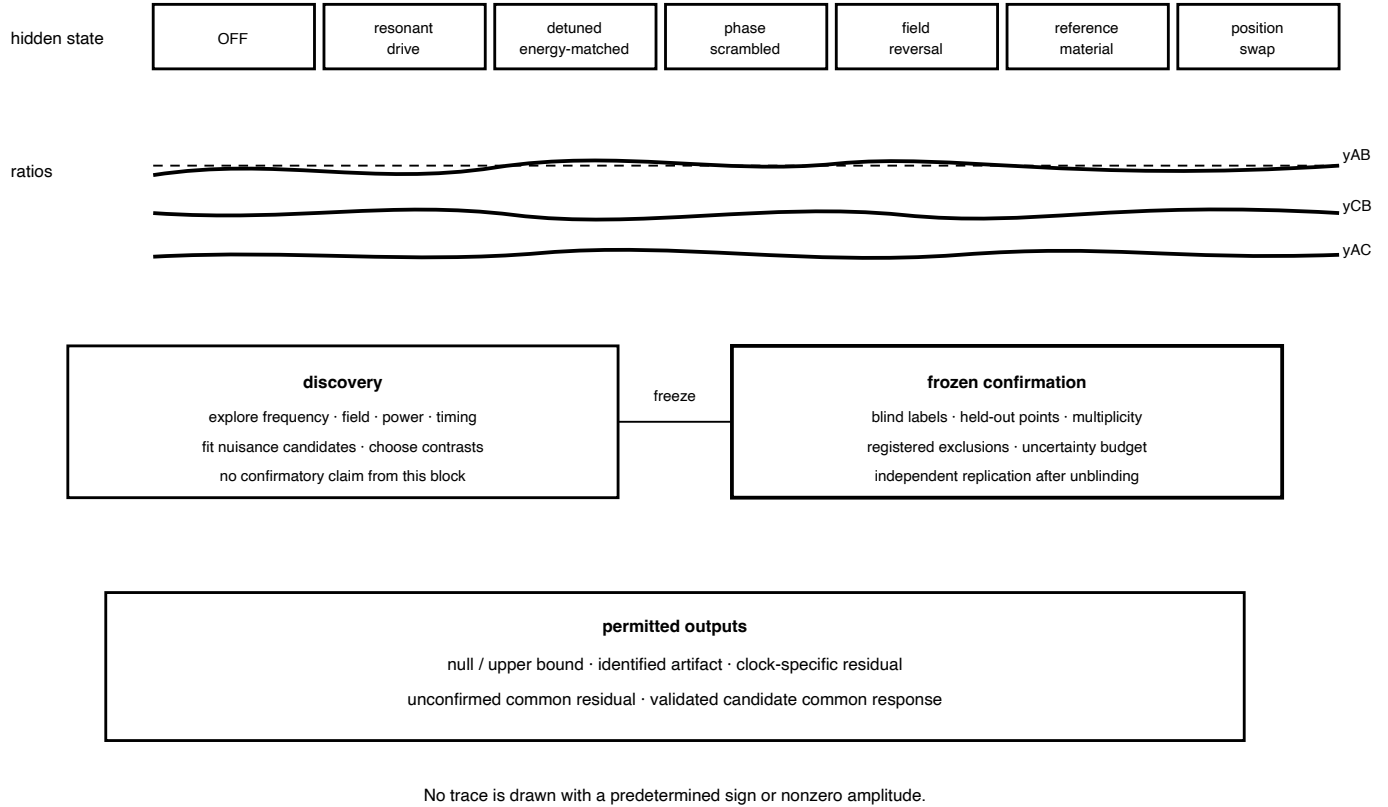


FIG. 9

Randomized states, discovery/confirmation separation, and a synchronous estimator whose allowed output includes a null or upper bound.

Joint clock and force channels with shared nuisance paths

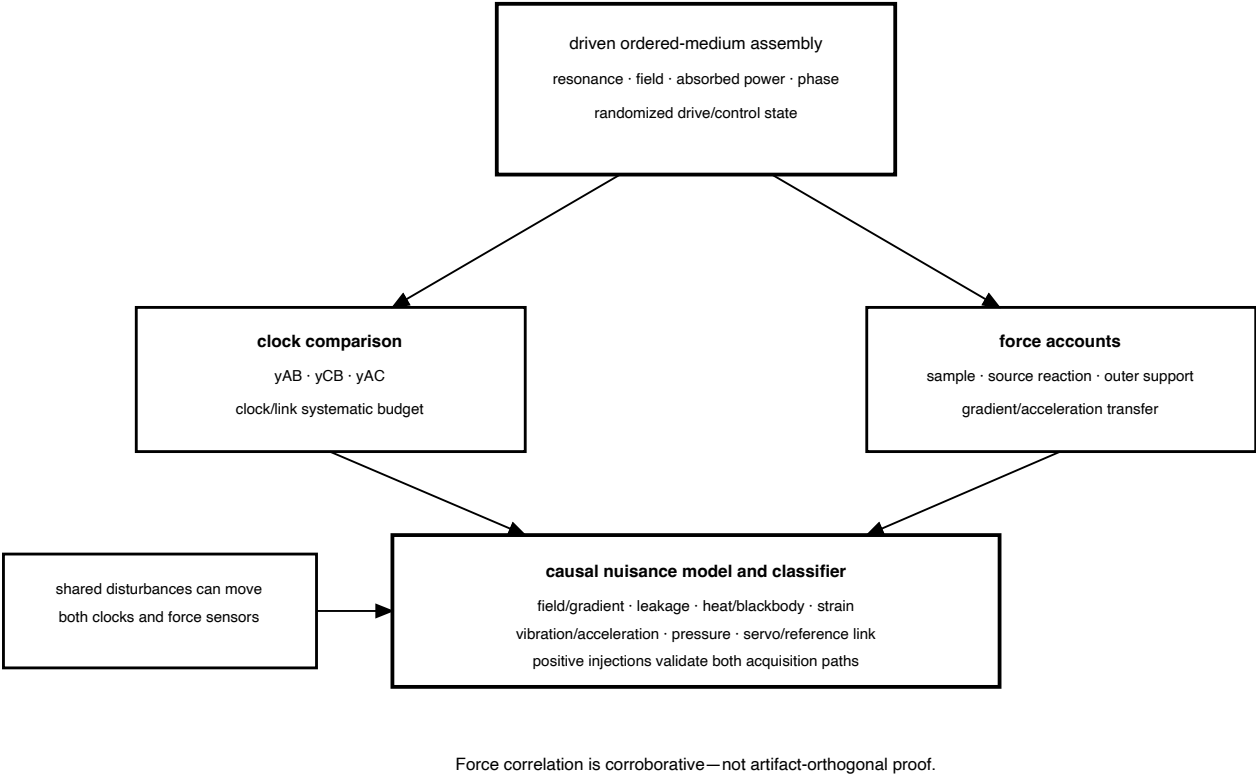
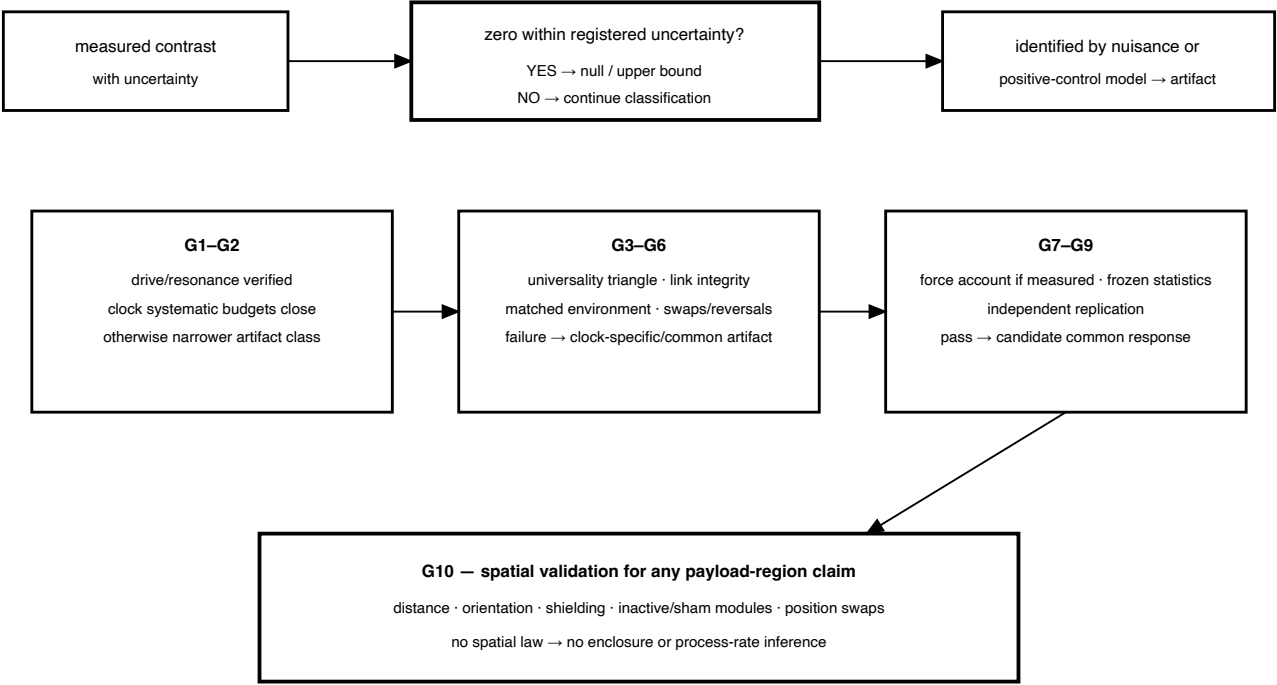


FIG. 10

Clock and force channels share ordinary disturbance pathways; simultaneous acquisition strengthens diagnosis only after the causal nuisance model and positive controls succeed.

Classification ladder and validation gates G1–G10

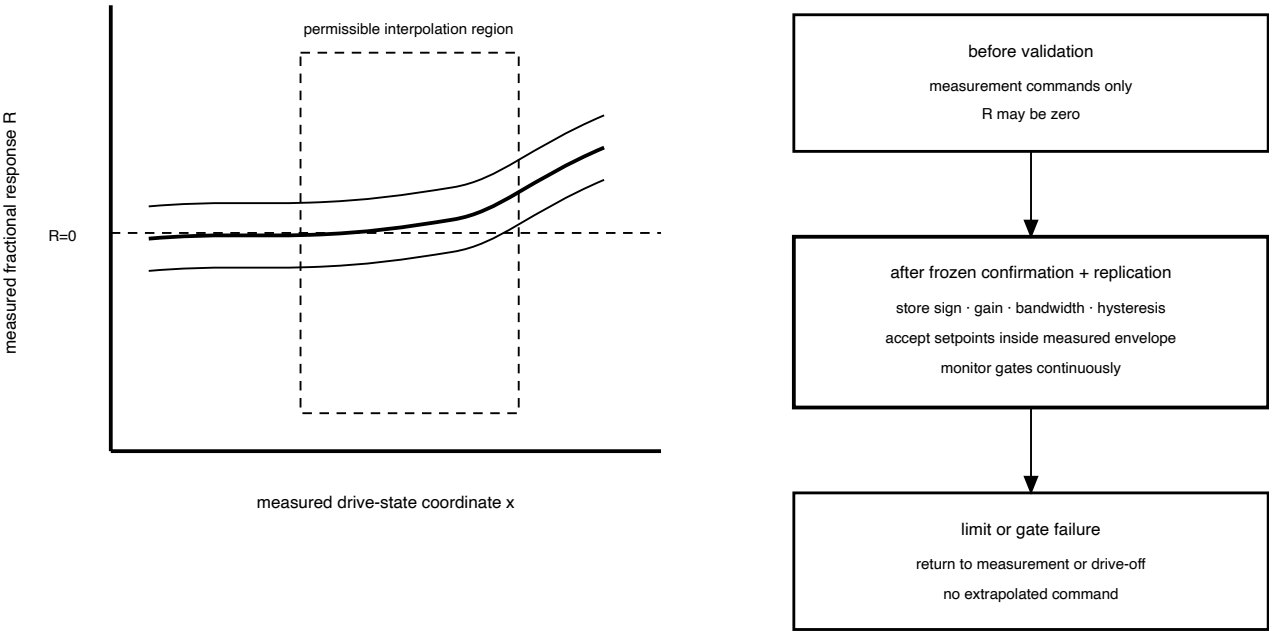


Each failed applicable gate returns the result to the narrowest supported class; null is complete.

FIG. 11

Classification and validation: the system earns progressively stronger labels through measured gates and never converts mere synchrony into a proper-time conclusion.

Empirical response map and conditional-control boundary

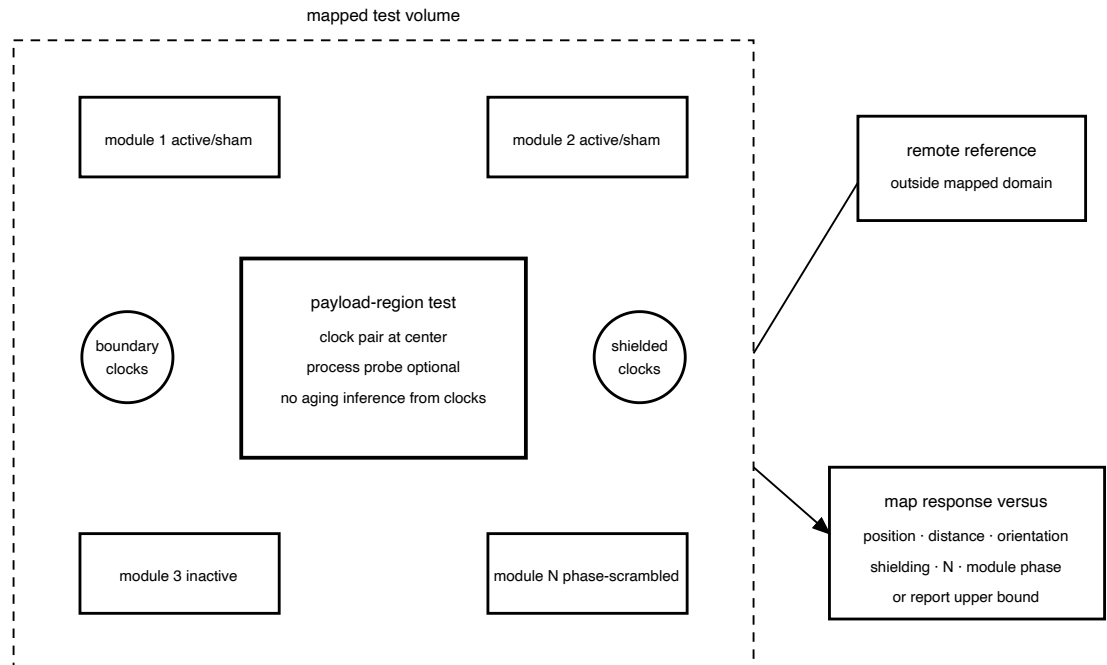


No fixed offset, sign, power law, or metrology-to-enclosure ladder is presumed.

FIG. 12

The empirical rate-response characteristic may be zero. The product setpoint mode unlocks only after validation and only inside the permissible interpolation region stored in the map.

Spatial and array qualification for the G4 enclosure—effect not presumed



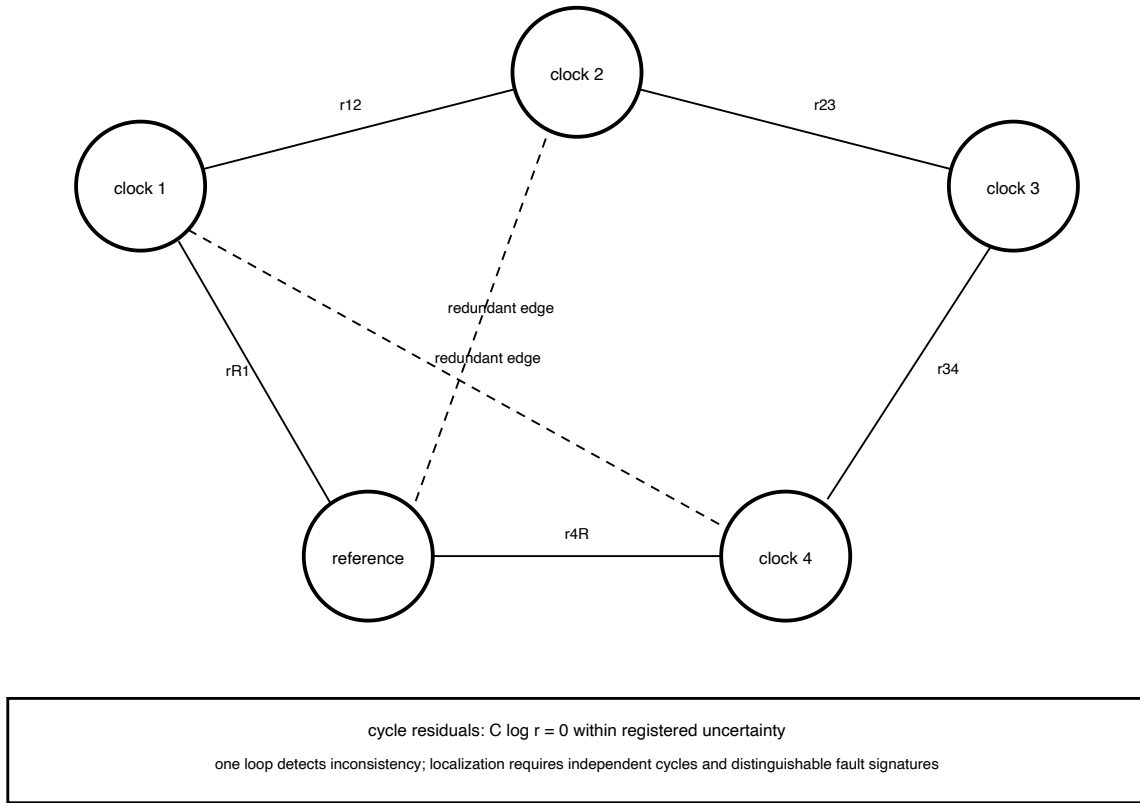
G10 must pass at a position before it belongs to the G4 payload region; the spatial and allocation maps are written here.

Sequential · simultaneous · inactive · sham · phase-scrambled · distance · shielding states

FIG. 13

Null-valid spatial mapping (E8, G10) qualifies the G4 payload-region enclosure. Multiple witness locations and controlled module states measure or bound reach and additivity; the payload region is the set of positions that pass.

Redundant comparator network and cycle-space fault test



This integrity test does not assert physical multiplication of actuator effects.

FIG. 14

Four or more nodes with sufficient links, or three nodes with independently realized parallel links, create plural independent cycle residuals. The controller reports ambiguous candidates unless the registered fault model is observable; in deployment the same test runs on the witness/reference network at the comparison port.